Chapter 5. Setting the Ventilator

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Setting the Ventilator: Introduction

The choice of ventilator settings should be guided by clearly defined therapeutic end points. In most instances, the primary goal of mechanical ventilation is to correct abnormalities in arterial blood-gas tensions. In most patients, this is accomplished easily by adjusting the minute volume to correct hypercapnia and by treating hypoxemia with oxygen (O_2) supplementation. Because the volume, frequency, and timing of gas delivered to the lungs have important disease-specific effects on cardiovascular and respiratory systems functions, the physician must avoid simply managing the blood-gas tensions of the ventilator-dependent patient. After a brief review of the capabilities of modern ventilators, this chapter discusses the mechanical determinants of patient–ventilator interactions and defines therapeutic end points in common respiratory failure syndromes. These sections provide background for the major thrust of the chapter, which is to detail the physiologic consequences of positive-pressure ventilation and to develop recommendations for ventilator settings in various disease states based on this knowledge.

Capabilities of Modern Ventilators

The incorporation of microprocessors into ventilator technology has made it possible to program ventilators to deliver gas with virtually any pressure or flow profile. Significant advances have been made in producing machines that are more responsive to changes in patient ventilatory demands, and most full-service mechanical ventilators display diagnostic information contained in airway pressure (Paw), volume (V), and flow ($\dot{V}$) waveforms. Because of these added capabilities, the practitioner is being challenged with a staggering array of descriptive acronyms for so-called new modes of ventilation. To avoid unnecessary confusion, it is useful not to focus on specific modes for the moment but rather to consider three general aspects of ventilator management: (a) the choice of inspired-gas composition, (b) the means to ensure the machine's sensing of the patient's demand, and (c) the definition of the machine's mechanical output.

Choice of Inspired-Gas Composition

Practically speaking, decisions regarding the composition of inspired gas concern only the O_2 concentration (see "Acute Lung Injury and Hypoxic Respiratory Failure" below). Although there may be occasions when the care provider considers supplementing the inspired gas with nitric oxide, the efficacy of nitric oxide therapy for most forms of hypoxic respiratory failure remains to be established.[1] There has been growing interest in the biologic effects of hypercapnia on gas exchange, vascular barrier properties, and innate immunity.[2–7] Therapeutic hypercapnia, that is, the deliberate supplementation of inspired gas with carbon dioxide (CO_2), however, cannot be recommended at this point in time. On extremely rare occasions, it may be appropriate to use a helium-oxygen mixture in an attempt to lower the flow resistance across a lesion in the distal trachea or mainstem bronchi, and there has been some interest in the use of helium in asthma.[8] Currently, these approaches must be considered experimental.

Machine's Sensing of Patient's Demand (Ventilator Triggering)

Ideally, a mechanical ventilator should adjust not only its rate but also its instantaneous mechanical output in response to changing patient demands. Conventional modes of ventilation cannot do so; instead, such modes execute a predefined pressure or flow program after an effort has been sensed.

Volume preset controlled mechanical ventilation (CMV) refers to a mode during which rate, tidal volume (V_T), inspiratory-to-expiratory timing (I:E ratio), and inspiratory flow profile are determined entirely by machine settings and cannot be altered by either the rate nor the amplitude of the patient's effort. Occasionally, investigators refer to ventilation as "controlled" when spontaneous respiratory muscle activity has been abolished by mechanical hyperventilation or by pharmacologic means (e.g., sedation and neuromuscular blockade).

Assist-control ventilation (ACV) gives the patient the option of initiating additional machine breaths when the rate, set by the physician, is insufficient to meet the patient's rate demand. ACV differs from intermittent mandatory ventilation (IMV) in that all delivered breaths execute the same pressure or flow program, depending on the choice of primary mode. The ACV feature has lured many providers into the erroneous assumption that the primary machine rate setting is unimportant (see "Acute Lung Injury and Hypoxic Respiratory Failure: Respiratory Rate" below).

Traditionally, machine algorithms for detecting patient effort have keyed on the airway pressure signal.[9] Because the inspiratory port of ventilators is closed during machine expiration, any inspiratory effort that is initiated near relaxation volume (Vrel) causes a fall in Paw. When Paw reaches a predefined trigger threshold (usually set 1 to 2 cm H_2O below the end-expiratory pressure setting), the machine switches from expiration to inspiration. In the presence of dynamic hyperinflation, the inspiratory muscles must generate considerably more pressure than the set airway trigger pressure before a machine breath is delivered[10] (see “Obstructive Lung Diseases” below).

Particularly in older ventilator models and in less-sophisticated portable machines intended for home use, it used to be common to find delays of up to 0.5 second between the onset of inspiratory muscle activity and machine response. In most ventilators used today, such delays are less than 100 milliseconds.[11] Sensing delays are common when the Paw is monitored in the machine rather than near the patient–ventilator interphase. In the former case, the ventilator tubing acts as a capacitor, delaying the transmission of pressure from the intrathoracic airway to the pressure transducer. Additional delays can be attributed to dynamic hyperinflation and physical constraints on the opening and closing of demands valves. Considering that most ventilator-dependent patients generate between 4 and 8 cm H_2O pressure in 100 milliseconds,[10,12] delays can cause significant effort expenditure and discomfort. More importantly, patients may terminate seemingly ineffective inspiratory efforts prematurely only to initiate another effort of greater amplitude shortly thereafter. This leads to discrepancies between patient and machine rate.[13,14] Discrepancies are seen often in weak or heavily sedated patients with severe hyperinflation and high intrinsic respiratory rates.[13,15]

Flow-triggering algorithms are alternatives or adjuncts to Paw-based triggering. During “flow triggering,” a base flow of gas is being delivered to the patient during the expiratory as well as the inspiratory phases of the machine cycle.[9] Unless the patient makes an inspiratory effort, gas bypasses the endotracheal tube and is discarded through the expiratory machine port. In the absence of patient effort, expiratory flow is equal to inspiratory base flow. In the presence of an inspiratory effort, gas enters the patient’s lungs and is thereby diverted from the expiratory machine port. A discrepancy between inspiratory and expiratory base flow is sensed, and the ventilator switches phase. Because “flow triggering” alleviates the need to rarefy gas against an occluded demand valve, initially it was considered superior to pressure-based trigger algorithms.[9,16] Because most new-generation ventilators have combined pressure and flow-sensing capabilities, these distinctions no longer apply.

Options for Defining the Machine’s Mechanical Output

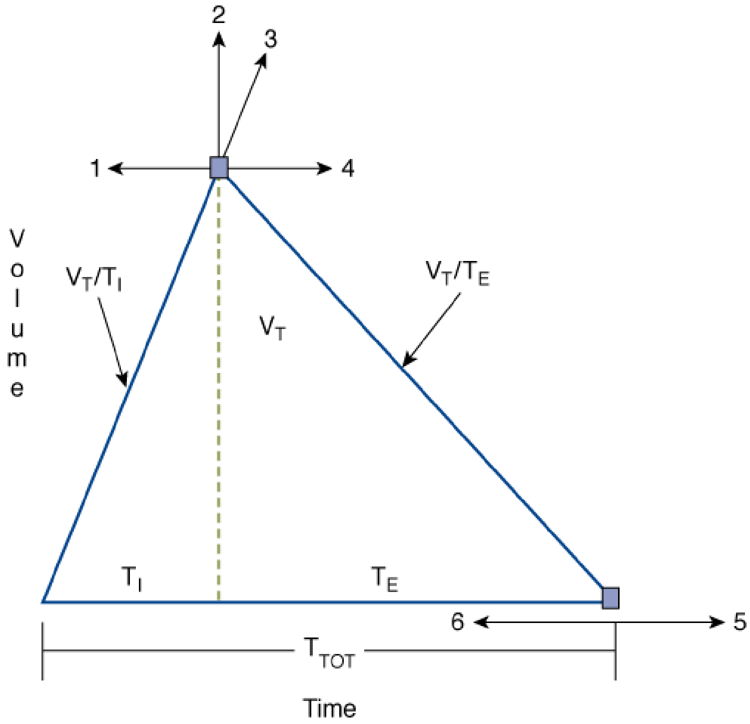
The mode of mechanical ventilation often refers to the shape of the inspiratory pressure or flow profile and determines whether a patient can augment V_T or rate through his or her own efforts.

Volume-Preset Mode

In conventional volume-preset mode, each machine breath is delivered with the same predefined inspiratory flow-time profile. Because the area under a flow-time curve defines volume, V_T remains fixed and is uninfluenced by the patient’s effort. Volume-preset ventilation with constant (square wave) or decelerating inspiratory flow is the most widely used breath-delivery mode. Breath delivery with flows that decrease with increasing lung volume are effective in reducing peak Paw. It is not clear, however, whether they protect the lungs from overdistension injury any more than square wave flow profiles.

The mechanical output of a ventilator operating in the volume-preset mode is uniquely defined by four settings: (a) the shape of the inspiratory flow profile, (b) V_T, (c) machine rate, and (d) a timing variable in the form of either the I:E ratio, the duty cycle (T_I/T_{TOT} [inspiratory time/total respiratory time]), or the T_I. In some ventilators, timing is set indirectly through the choice of peak or mean inspiratory flow (V_T/T_I). Figure 5-1 illustrates the relationships among these and other breathing-pattern parameters of significance.

Figure 5-1



Idealized spirogram of a breath delivered during volume-preset mechanical ventilation. Examples 1 through 6 indicate specific changes in ventilator settings and illustrate the consequences on flow and timing variable. (For abbreviations, see Table 5-1.) (1) Increasing mean inspiratory flow (V_T/T_I) at a constant machine rate setting results in a reduced I:E ratio and vice versa. T_I, T_I/T_{TOT}, and mean expiratory flow (V_T/T_E) decline. (2) Increasing V_T at constant T_I/T_{TOT} or I:E setting increases mean inspiratory flow and requires an increase in mean expiratory flow. Remember that mean inspiratory flow equals peak inspiratory flow when delivery modes with constant square wave flow profiles are used. (3) Increasing V_T at a constant mean inspiratory flow setting increases T_I, T_I/T_{TOT}, I:E ratio, and mean expiratory flow. (4) Decreasing mean inspiratory flow at a constant machine rate setting results in an increase in the I:E ratio and vice versa. T_I, T_I/T_{TOT} and mean expiratory flow rise. (5) Reducing the machine backup rate (f_M) at a fixed I:E ratio or T_I/T_{TOT} setting always prolongs TI and lowers mean inspiratory flow. The timing effects of reducing f_M at a fixed inspiratory-flow setting cannot be predicted without knowledge of the patient's actual trigger rate. (6) Increasing f_M at a fixed I:E ratio or T_I/T_{TOT} setting always raises inspiratory flow. The timing effects of increasing f_M at a fixed inspiratory-flow setting cannot be predicted without knowledge of the patient's actual trigger rate.

Table 5-1: List of Abbreviations

τ	Time constant
AC	Assist-control mode
ARDS	Adult respiratory distress syndrome
CMV	Controlled mechanical ventilation
CPAP	Continuous positive airway pressure
Ers	Elastance of the respiratory system
Edi	Electromyographic tracing of the diaphragm
F	Force
f_A	Actual breathing rate
$FEF_{25–75}$	Forced expiratory flow in the mid vital capacity range
FI_{O_2}	Fractional inspired oxygen concentration
f_M	Machine backup rate
I:E ratio	Inspiratory-to-expiratory time ratio
IMV	Intermittent mandatory ventilation
Pa_{CO_2}	Arterial CO_2 tension
Pa_{O_2}	Arterial O_2 tension
Paw	Airway pressure
PCV	Pressure-controlled ventilation
PEEP	Positive end-expiratory pressure
$PEEP_E$	Extrinsic positive end-expiratory pressure
PEEPi	Intrinsic positive end-expiratory pressure
Pel	Elastic recoil pressure
P_{tp}	Transpulmonary pressure
Pmus	Inflation pressure exerted by inspiratory muscles
Pres	Resistive pressure
Prs	Recoil of respiratory system
PSV	Pressure support ventilation
SIMV	Synchronized intermittent mandatory ventilation

T_E	Expiratory time
T_I	Inspiratory time
T_I/T_{TOT}	Duty cycle
T_{TOT}	Total cycle time
TLC	Total lung capacity
$\dot{V}$	Flow
$\dot{V}/\dot{Q}$	Ventilation-perfusion ratio
$\dot{V}_{CO_2}$	Volume of CO_2 produced in liters per minute
$\dot{V}_E$	Minute ventilation
$\dot{V}_1$	Mean inspiratory flow
V(t)	Instantaneous lung volume
$\dot{V}_E/V_T$	Dead-space-to-tidal-volume ratio
Vee	Volume of lungs at end expiration
Vrel	Relaxation volume
V_T/T_E	Mean expiratory flow
V_T/T_I	Mean inspiratory flow
V_T	Tidal volume
Vtrapped	Volume of gas remaining in the elastic element at the beginning of a new machine inflation
Wel	Elastic work

Pressure-Preset Mode

During pressure-preset ventilation, the ventilator applies a predefined target pressure to the endotracheal tube during inspiration. The resulting V_T and inspiratory flow profile varies with the impedance of the respiratory system and with the strength and duration of the patient's inspiratory efforts. Therefore, when the lungs or chest wall become stiff, airway resistance increases, the patient's own inspiratory efforts decline, or T_I decreases, V_T decreases. An increase in respiratory system impedance can lead to a dangerous fall in minute ventilation ($\dot{V}_E$), hypoxemia, and CO_2 retention, but in contrast to volume-preset modes, it does not predispose the patient to an increased risk of barotrauma. On the other hand, pressure-preset modes are no safeguard against ventilator-induced lung injury because large fluctuations in respiratory impedance or patient effort would result in large V_T fluctuations directly undermining the primary objective of lung-protective mechanical ventilation (see section on Acute Lung Injury and Hypoxic Respiratory Failure).

Pressure-support ventilation (PSV), pressure-controlled ventilation (PCV) and airway pressure release ventilation (APRV) are the most widely used forms of pressure-preset ventilation. In contrast to PCV, PSV requires the patient's effort before a machine breath is delivered. Consequently, PSV is not suitable for the management of patients with central apneas. During PCV, the physician sets the machine rate, the T_I, and thus the I:E ratio. In PSV, phase switching is linked to inspiratory flow, which, in turn, depends on the impedance of the respiratory system, as well as on the timing and magnitude of inspiratory muscle pressure output.[11,14] APRV is akin to PCV with a long duty cycle, but with one important distinction: patients are able to take spontaneous breaths throughout all phases of the machine cycle.

PSV remains a popular weaning mode for adults. Its popularity is based on the premise that weaning from mechanical ventilation should be a gradual process and that the work of unassisted breathing through an endotracheal tube is unreasonably high and could lead to respiratory muscle failure in susceptible patients. Actual measurements of pulmonary resistance and work of breathing before and after extubation do not support this reasoning,[17,18] and several large clinical trials have established equivalence between PSV and T-piece weaning (unassisted breathing from a bias-flow circuit).[19–21] In the PSV mode, a target pressure is applied to the endotracheal tube, which augments the inflation pressure exerted by the inspiratory muscles (Pmus) on the respiratory system. When inspiratory muscles cease to contract and Pmus falls, inspiratory flow (a ventilator-sensed variable) declines, and the machine switches to expiration. Early PSV modules were designed to generate pressure ramps (square wave inflation pressure) and had relatively rigid flow-based, off-switch criteria. Most recent versions of PSV afford control over the rate of rise in inspiratory pressure and the flow threshold at which inspiration is terminated.[11,14,22]

Synchronized Intermittent Mandatory Ventilation

During synchronized intermittent mandatory ventilation, a specified number of usually volume-preset breaths are delivered every minute. In addition, the patient is free to breathe spontaneously between machine breaths from a reservoir or to take breaths augmented with PSV. Most ventilators allow the operator to choose between volume- and pressure-preset mandatory breaths. Unless the patient fails to breathe spontaneously, machine breaths are delivered only after the ventilator has recognized the patient's effort; that is, ventilator and respiratory muscle activities are "synchronized." Because nowadays all IMV circuits are synchronized, the terms *IMV* and *synchronized intermittent mandatory ventilation* are used interchangeably. Although synchronized intermittent mandatory ventilation remains a viable and popular mode of mechanical ventilation, compared with the alternatives, PSV and T piece, it has clearly proven inferior as a weaning modality.[19,20,23,24] Moreover, the care provider needs to be aware of certain pitfalls when using IMV. Even a small number of volume-preset IMV breaths per minute may make the blood-gas tensions look acceptable in patients who otherwise meet criteria for respiratory failure. One should suspect this in patients with small spontaneous V_T (≤3 mL/kg of body weight), in those with thirty or more inspiratory efforts per minute regardless of whether they trigger a machine breath, and when dyspnea and thoracoabdominal paradox indicate a heightened respiratory effort. One reason that IMV remains popular is because it silences apnea alarms by masking PSV-induced respiratory dysrhythmias, which are common in sleeping and obtunded patients.[25–27]

Dual-Control and Advanced Closed-Loop Modes

Many new-generation mechanical ventilators feature modes with closed-loop feedback control of both pressure and volume.[28,29] While a detailed description of the operating principles of every new mode is beyond the scope of this chapter, it is important to understand the rationale behind dual-control modes and some of their general operating characteristics. The idea behind most dual-control modes is the meeting of a ventilation target while maintaining low inflation pressures. To this end, ventilator output is adjusted based on volume, flow, and pressure feedback. This may occur within each machine cycle or gradually from one cycle to the next. Modes that adjust output within each cycle execute a predetermined pressure-time program as long as the desired V_T is reached. When the V_T target is not reached, inspiration continues at a preselected inspiratory flow rate (volume-limited) until the target volume is attained. *Volume-assured pressure support* and *pressure augmentation* are examples of such modes.[30] Breath-to-breath dual-control modes are pressure-limited and time-cycled or flow-cycled. Ventilator output is derived from the pressure–volume relationship of the preceding breath and is adjusted within predefined pressure limits to maintain the target V_T. *Adaptive support ventilation, pressure-regulated volume control, volume control+, autoflow, adaptive pressure ventilation, volume support,* and *variable pressure support* are examples of breath-to-breath control modes. There is no evidence that the use of dual-control modes improves patient outcomes.[31,32] Moreover, there is a conceptual problem insofar as less complex modes already safeguard against hypoventilation, whereas dual-control modes do little to protect the patient from a potentially harmful increase in the regulated variable, that is, large V_T-mediated lung injury.[33,34]

Neurally adjusted ventilatory assistance (NAVA) and *proportional-assist ventilation* (PAV)[35–37] are the most complex, and arguably the most promising, closed-loop ventilation modes. During NAVA, the diaphragm's electrical activity is recorded with an esophageal probe, and the signal is conditioned and transposed into a positive airway pressure output. During PAV, the ventilator derives its mechanical output from continuously monitored Paw, V, and $\dot{V}$ information, which, in turn, reflects Pmus. The operating principles of PAV will be easier to understand after a review of patient–ventilator interactions (see "The Mechanical Determinants of Patient-Ventilator Interactions" below). Compared to conventional modes of mechanical ventilation, both NAVA and PAV preserve the biologic variability in breathing rate and VT, which is generally considered lung protective.[38] Moreover, the maintenance of intrinsic respiratory control mechanisms is likely to reduce the probability of exposing the lungs to injurious deformations. Although there is ample literature on the effects of closed-loop modes on patient–ventilator interactions and physiologic end points, there is insufficient clinical experience to judge efficacy of these modes compared to conventional approaches. This is particularly true for NAVA, which was only recently introduced to the world market, but holds particular promise as platform for delivering noninvasive mechanical ventilation and as a support mode for neonates and small infants. Be this as it may, the full-scale migration of closed-loop modes from expert hands into general practice will likely depend on the willingness of providers to acquire the physiologic insights and skills necessary for managing patients who are ventilated with “unconventional” modes.

The Mechanical Determinants of Patient–Ventilator Interactions

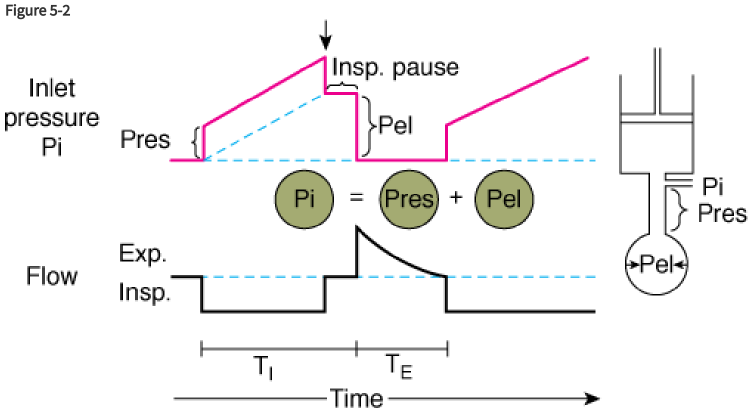
Inspiratory Mechanics

It is useful to think of patient–ventilator interactions in terms of a mechanical or electrical analog system consisting of a resistive element (resistor) and an elastic element (capacitor) in series. The forcing function is defined by the pressure or flow “program” that is executed by the mechanical ventilator.

In Figure 5-2, a piston pump (the mechanical ventilator) is attached to a rigid tube (the resistive element) and a balloon (the elastic element). An in-series mechanical arrangement means that at any time *t,* the pressure that is applied to the tube inlet Pi(*t*) (near the attachment to the ventilator) is equal to the sum of two pressures, an elastic pressure Pel(*t*) and a resistive pressure Pres(*t*):

$$Pi(t) = Pel(t) + Pres(t) \qquad (1)$$

Figure 5-2



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Components of inlet pressure. The model of the respiratory system at right consists of a resistive element (straight tube) and an elastic element (balloon) connected to a ventilator (piston). During inflation of the model with constant flow *(lower panel*), there is a stepwise increase inlet pressure (Pi) that equals the loss of pressure across the resistive element (Pres) (*upper panel*). Thereafter, Pi increases linearly and reflects the mechanical properties of the elastic element (Pel). Pi is the sum of Pres and Pel. At end inspiration, when flow has ceased (Insp. Pause), Pi decreases by an amount equal to Pres; Pi equals Pel during Insp. Pause. T_I, inspiratory time; T_E, expiratory time. (Used, with permission, from Hubmayr, et al. Physiologic approach to mechanical ventilation. *Crit Care Med.* 1990;18:103–113.)

The tube outlet pressure at the junction with the balloon is equal to the pressure inside the balloon, that is, Pel. Pres is the difference in pressure between the tube inlet and the tube outlet. Assuming linear-system behavior, the inlet pressure-time profile can be computed for any piston stroke volume (Vstroke) and flow ($\dot{V}$) setting, provided the resistive properties of the tube (R) and the elastic properties of the balloon (E) are known:

$$Pi(t) = EV(t) + R\dot{V}(t) \qquad (2)$$

The elastance (E) is a measure of balloon stiffness and is equal to the Pel-to-V stroke ratio (assuming 0 volume and pressure at the beginning of balloon inflation). Therefore, *Pel(t)* in Eq. (1) can be replaced with *EV(t)* in Eq. (2). Because the Ohm law states that the tube resistance (R) is equal to the Pres-to-$\dot{V}$ ratio, *Pres(t)* in Eq. (1) can be replaced with the product $R\dot{V}(t)$ in Eq. (2).

Equations (1) and (2) are based on the equation of motion, which describes the force (F) that must be applied to a mass (M) in order to move it a certain distance (d) at a rate *dd/dt* against a spring (elastic) load:

$$F(t) = kd(t) + k'(dd/dt)(t) + k''[d(dd/dt)](t) \qquad (3)$$

where k = stiffness of the spring (analogous to E); k′ = frictional resistance between mass and supporting surface (analogous to R); and k″ = inertance, which is proportional to mass.

The first and second derivatives of *d* (analogous to volume) represent the velocity (*dd/dt,* analogous to flow) and the acceleration [*d*(*dd/dt*)/*dt*] of the mass at time *t,* respectively. As long as the mass of the moving parts in the model of Figure 5-2 is small, any inertive-pressure component that is dissipated during the acceleration of gas at the beginning of the pump instroke can be ignored. Therefore, the respiratory analog of the equation of motion [Eqs. (1) and (2)] considers only elastic and resistive pressures.

Consider the Pi-time profile of a tube-balloon system with resistance of 10 cm H_2O × L/s and an elastance of 10 cm H_2O when the piston pump is programmed to deliver a volume of 0.5 L with a constant (square wave) flow of 0.5 L/s. Because flow and R remain constant throughout inflation, Pres is constant at 5 cm H_2O and accounts for the initial step change in inlet pressure at the beginning of inflation. As gas enters the balloon, Pi increases further and reaches a value of 10 cm H_2O at end inflation. At that instant, the tube is occluded (end-inflation hold), causing Pi to drop by an amount equal to Pres (as flow returns to 0). The end-inflation hold pressure is equal to Pel at that volume. Its value of 5 cm H_2O is equal to the product of piston stroke volume (0.5 L) and elastance (10 cm H_2O/L), as follows from Eqs. (1) and (2). Subtracting Pres from Pi(*t*) yields the Pel per time course during inflation. Pel increases linearly with time and volume. Its rate of rise (*d*Pel/*dt*) parallels that of Pi and is determined by E and the flow setting of the piston:[39]

$$E \times \dot{V} = (dp/dV) \times (dV/dt) = dP/dt \qquad (4)$$

Although changes in inspiratory flow result in proportional changes in *d*Pel/*dt,* flow has no effect on peak Pel, provided that Vstroke, and thus peak lung volume, is held constant. This is in contrast to peak Pi, which reflects flow-dependent changes in Pres, as well as change in Pel, at end inflation. The relevance of this important property of linear single-compartment systems will become apparent later when the relationships between ventilator settings and barotrauma (balloon yield stress) are discussed (see “Acute Lung Injury and Hypoxic Respiratory Failure” below).

Expiratory Mechanics

In mechanically ventilated subjects, expiration is usually a passive process that is driven by the elastic recoil (*Pel*) of the respiratory system. Assuming linear pressure-volume and pressure-flow relationships, the instantaneous expiratory flow [$\dot{V}exp(t)$] is given by

$$\dot{V}exp(t) = Pel(t)/R \qquad (5)$$

Because *Pel(t)* is a function of *E* and of the instantaneous lung volume [V(*t*)], Eq. (5) can be rewritten as

$$\dot{V}exp(t) = E \times V(t)/R = V(t)/R \times C \qquad (6)$$

where *C* (the compliance of the respiratory system) is simply the inverse of the elastance (*E*). The product of *R* and *C* characterizes the time constant (τ) of single-compartment linear systems. The time constant defines the time at which approximately two-thirds of the volume above Vrel has emptied passively. From this it should be clear that patients with increased respiratory system resistances and compliances (e.g., patients with emphysema) are prone to dynamic hyperinflation even if one ignores nonlinear system behavior, such as flow limitation, for the moment.

The volume of gas remaining in the elastic element at the beginning of a new machine inflation (*Vtrapped*) can be calculated as follows:

$$Vtrapped = V_T/(e^{TE/\tau} - 1) \qquad (7)$$

In other words, the degree of dynamic hyperinflation is determined by the choice of ventilator settings, specifically mean expiratory flow (V_T/V_E) and the time constant of the respiratory system, which reflects its mechanical constants *R* and *C.*[40] These important concepts are expanded on under “Obstructive Lung Diseases” below.

Limitations of Linear Single-Compartment Models

Before linear model principles are applied to the ventilator management of patients, one must be cognizant of the model’s limitations. The limitations fall into two general categories: those related to nonlinear respiratory system characteristics and those related to respiratory muscle activation during mechanical ventilation. Sources of nonlinear system behavior include inhomogeneities within the numerous bronchoalveolar compartments (particularly when the lungs are diseased),[41] respiratory system hysteresis from recruitment of alveolar units and time-dependent surface tension phenomena,[42] and phenomena related to dynamic airway collapse and expiratory flow limitation.[43] Coactivation of the respiratory muscles during mechanical ventilation invalidates Eqs. (1) through (7) insofar as they alter the impedance of the respiratory system and change the driving pressure for expiratory flow. If one assumes that the respiratory muscles and the ventilator are arranged in series, then the monitoring of pressure, volume, and flow at the airway opening offers the opportunity to define the magnitude, rate, and duration of respiratory muscle output in mechanically ventilated subjects.[44,45]

Defining Therapeutic End Points in Common Respiratory Failure Syndromes

Numerous diseases of cardiopulmonary systems can cause respiratory failure. From a ventilator management perspective, it is useful to group them into those that cause lung failure and those that cause ventilatory pump failure.

The hallmark of lung failure is hypoxemia, which is usually the result of severe ventilation-perfusion mismatch. The hallmark of ventilatory pump failure is hypercapnia. Ventilatory pump failure may be caused by disorders of the central nervous system, peripheral nerves, or respiratory muscles. It also may accompany diseases of the lungs, such as emphysema, once the ventilatory pump fails to compensate for inefficiencies in pulmonary CO_2 elimination. Two classic examples of hypoxic and hypercapnic ventilatory failure that require fundamentally different approaches to mechanical ventilation are the acute respiratory distress syndrome (ARDS) and chronic airflow obstruction. The therapeutic goal in ARDS is to protect the lung from mechanical injury while raising lung volume in an attempt to reduce shunt by reexpanding collapsed and flooded alveoli. In contrast, the therapeutic goal in a patient with hypercapnic ventilatory failure from exacerbation of airways obstruction is to reduce dynamic hyperinflation and to protect the respiratory muscles from overuse.

Acute Lung Injury and Hypoxic Respiratory Failure

Acute lung injury (ALI) is a syndrome associated with bilateral pulmonary infiltrates and a gas-exchange impairment severe enough to lower the arterial oxygen tension-to-fractional inspired oxygen concentration ratio (Pa_{O_2}/FI_{O_2}) below 300.[46] Heart failure and moderate to severe preexisting chronic lung disease must be absent. ALI and its more severe form, ARDS, are often complications of systemic illnesses such as sepsis.[47] The impairment on pulmonary gas exchange therefore is accompanied frequently by microcirculatory failure. The general management goal in these disorders is to augment systemic oxygen delivery until the metabolic demands of the organism can be met. This goal requires an integrated approach between cardiovascular and ventilator support.[48]

Ventilator support is often difficult because exceedingly high ventilatory requirements challenge the performance capacity of mechanical ventilators; render patients at risk for barotrauma, ventilator-induced lung injury, and cardiovascular collapse; and often are accompanied by excessive respiratory muscle activity ("fighting the ventilator"). All these conditions on occasion can necessitate heavy sedation and neuromuscular blockade.

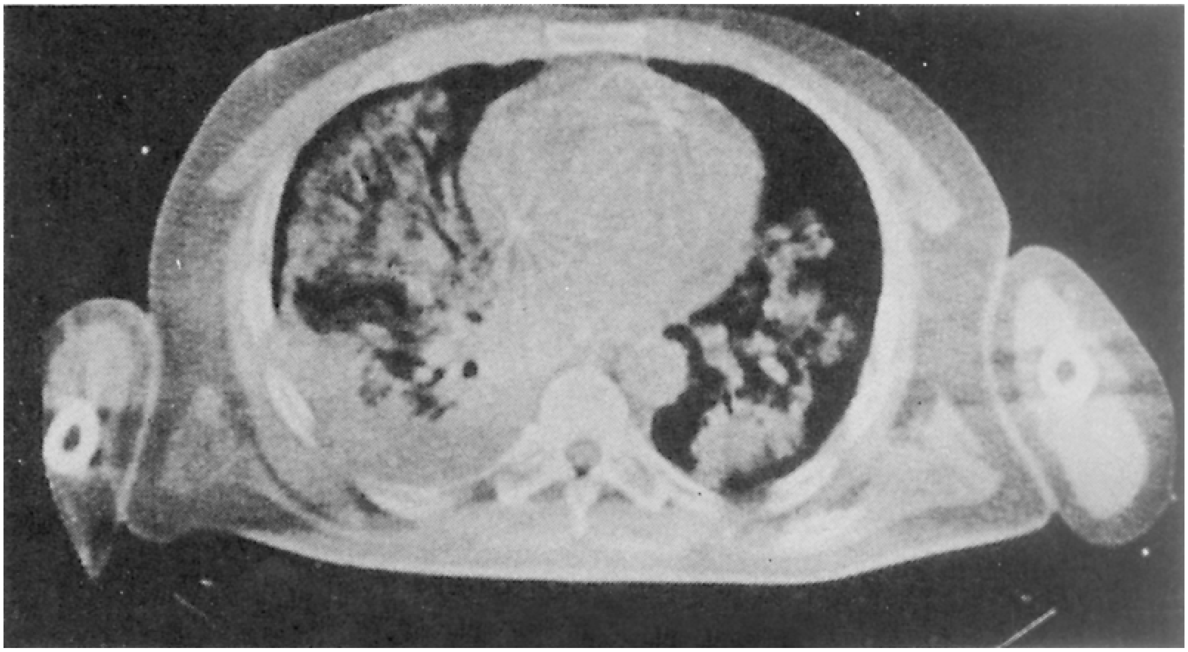
Fractional Inspired Oxygen Concentration

The two principal means by which the physician can increase Pa_{O_2} in ARDS are to raise the FI_{O_2} and to elevate the volume about which the lungs are being ventilated. The danger inherent in raising FI_{O_2} is oxygen toxicity,[49] whereas manipulating lung and/or V_T may result in ventilator-induced lung injury[34,50] and/or barotrauma.[51] Presented with the choice between two different kinds of adverse reactions, physicians currently tend to be more fearful of mechanical lung injury than of oxygen toxicity. Unfortunately, there are no clinical outcome studies that shed light on the interactions between these two iatrogenic insults. It is common practice to initiate ventilator support with an FI_{O_2} of 1.0 and to ignore the potential for oxygen toxicity during the first few hours of ventilator management. Although, generally speaking, the relative and combined risks of oxygen toxicity and overdistension injury of the lungs remain poorly defined, there are instances in which it seems wise to minimize FI_{O_2}, namely, in patients who have received drugs such as bleomycin or amiodarone, which make the lungs particularly susceptible to reactive O_2 species-mediated injury.[52]

Manipulating End-Expiratory Lung Volume

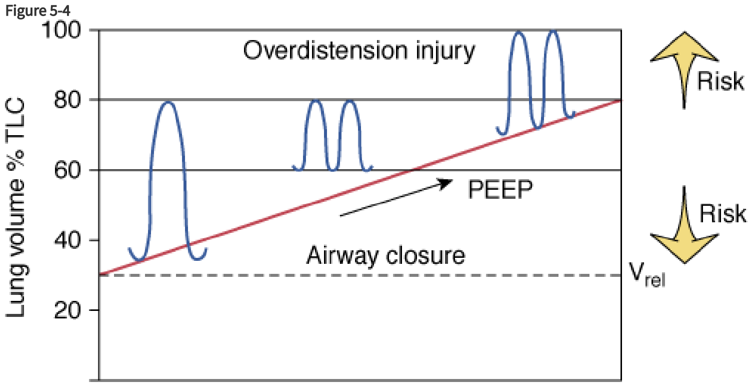
The insults to the lungs of patients with hypoxic respiratory failure are often patchy and result in flooding and closure of dependent airspaces (Fig. 5-3).[53,54] Paraspinal regions of the lung appear most susceptible to atelectasis (lack of aeration) because, in the supine posture, they normally empty close to their residual volume and they receive most of the pulmonary blood flow.[55] Therefore, insults to their capillary integrity are most likely to promote alveolar flooding, closure of airspaces by liquid plugs, surfactant inactivation, and gas-absorption atelectasis.[56] The accumulation of airway liquid and foam also generates interfacial forces that are large enough to abrade the epithelial lining of small airways during breathing, causing further injury.[57–61] Ventilator management therefore must be directed toward preventing the repeated opening and closing of unstable lung units, which means reestablishing aeration and ventilation of the flooded lung (Fig. 5-4).[62]

Figure 5-3



Computed tomographic (CT) scan of a patient with acute respiratory failure in the supine position. Note the patchy, nonuniform distribution of alveolar edema. (Used, with permission, from Gattinoni L, et al. Body position changes redistribute lung computed-tomographic density in patients with acute respiratory failure. *Anesthesiology.* 1991;74:15–23.)

Figure 5-4



Schematic of the therapeutic end points of ventilator management in ARDS. Raising peak lung volume toward total lung capacity (TLC) increases the risk of barotrauma. Keeping the minimal volume near relaxation volume (Vrel) raises the likelihood of alveolar derecruitment at end expiration and the need to apply large opening pressures so as to recruit collapsed and flooded lung regions during the subsequent breath.

There are several ways to achieve this objective: (a) by raising overall lung volume through the judicious use of extrinsic positive end-expiratory pressure (PEEP), (b) by raising lung volume dynamically through "intentional gas trapping," (c) by increasing V_T, and (d) by taking advantage of the local distending forces generated by an actively contracting diaphragm. Any one of these approaches may be combined with so-called recruitment maneuvers, which consist of sustained (up to 40 seconds) inflations of the lungs to high volumes and pressures.[63–65] The preferred and time-tested approach is the judicious use of extrinsic PEEP. All the other means of raising lung volume are comparatively untested, and in the case of V_T, manipulations can be outright harmful.[33,34] Although there is a strong physiologic rationale to condition (i.e., "open") the lungs with recruitment maneuvers before a PEEP adjustment, most experimental and clinical data indicate that conditioning effects are relatively short-lived.[66–69] Because it is common for patients with ALI to have an increased respiratory rate, a component of dynamic hyperinflation is often present.[70] Despite the short time constant for lung emptying, the use of extrinsic PEEP valves, which in older-generation ventilators represent resistive as well as threshold loads, and ventilator settings that require large mean expiratory flows (V_T/T_E; see "Mean Expiratory Flow: The Hidden Variable" below) contribute to dynamic hyperinflation.

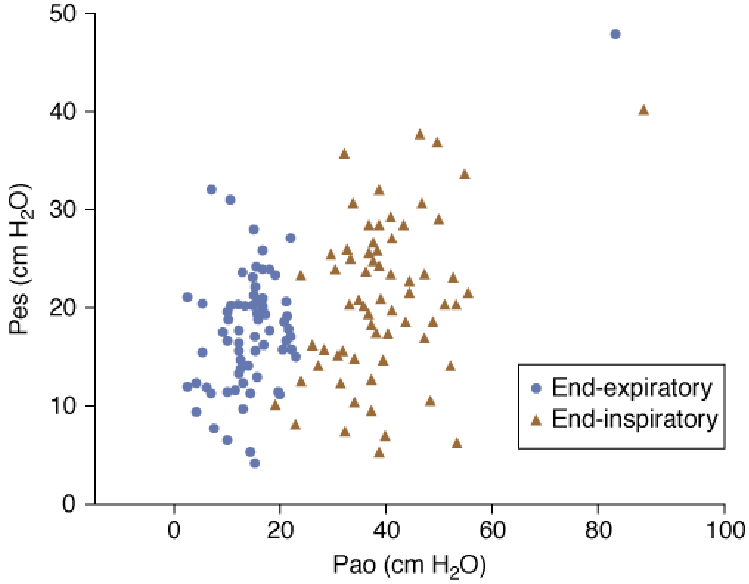
Although the experimental evidence in support of PEEP therapy in injured lungs is overwhelming, its specific application to clinical practice remains controversial. There is general agreement among experts that patients with injured lungs should be ventilated with PEEP settings greater than 5 cm H_2O. The risk, however, of overinflating and thereby damaging well-aerated, generally nondependent lung units and adverse hemodynamic effects set limits to an aggressive recruitment strategy.[71,72] Uncertainty about the topographic distribution of lung parenchymal stress and related stress injury thresholds are partly the reason why there is no consensus as to whether PEEP should be set arbitrarily to 10, 15, or 20 cm H_2O, whether it should be targeted to specific physiologic end points, and, if so, what those end points and their specific target thresholds should be. Several large randomized clinical trials have failed to resolve the controversy about "best PEEP."[73–75] Although none of these trials established superiority of one specific PEEP strategy over another, proponents of aggressive lung recruitment argue that PEEP was not targeted to the appropriate surrogate end points. Specifically, lung recruitment, chest wall recoil, and parenchymal stress were not measured or considered in the choice of PEEP settings.

Tools for assessing recruitment responses include (a) measures of regional lung aeration with computed tomography or electrical impedance imaging of the chest,[76–78] (b) measurement of lung and/or respiratory system pressure–volume relationships,[79–82] and (c) assessment of within-breath oscillations in arterial O_2 tension with indwelling arterial O_2 sensors with fast response times.[83,84] At the bedside, the most readily available PEEP management guides are airway inflation pressure amplitude, ΔP (in case of volume preset ventilation) or V_T (in case of pressure preset ventilation). As long as raising PEEP causes recruitment of previously "closed" lung units without overdistending already open ones, ΔP will decrease, reflecting the corresponding increase in compliance. In relaxed patients who are being ventilated with a pressure preset mode, the PEEP-related effect on lung compliance can be inferred from corresponding V_T changes. Adjusting PEEP until ΔP reaches a minimum, or conversely in the case of pressure preset ventilation until V_T reaches a maximum, is in line with the stress-index hypothesis.[81] The latter states that inflating lungs over the linear range of the respiratory system pressure–volume curve is most lung protective.

Patients who are likely to recruit in response to PEEP and who indeed may benefit from raising PEEP above 10 cm H_2O at the outset are patients with an increased end-expiratory chest-wall recoil pressure, which may or may not be associated with a reduced chest-wall/abdominal compliance[85–87] and patients whose airway and alveolar edema can be redistributed easily.[88,89] In critically ill patients, the most common conditions associated with increased chest-wall recoil are obesity, ileus, and ascites.[84,90] The ability to influence the distribution of edema within and between lung regions is greatest in the early stages of inflammation. In the later stages of ARDS, when the inflammatory exudate turns from liquid to a gel, it becomes much harder to "open" a closed airspace. The likelihood of high PEEP causing recruitment is even less once organizing pneumonia, alveolar remodeling, and fibrosis dominate the pathology.[91,92]

One attempt to identify groups of patients who are more or less likely to respond to PEEP has been to classify their insults as indirect versus direct.[89] Indirect insults such as abdominal sepsis are more likely associated with a favorable PEEP response (possibly because their chest-wall recoil is high and their alveolar exudate is liquid), whereas a direct insult, from a microbial lung infection, for example, tends to be more PEEP resistant (airway secretions tend to be viscous, and the alveolar exudate has the consistency of a gel). There is, however, enough variability in lung and chest-wall mechanics within and across these two patient populations to warrant a case-by-case assessment of pulmonary and hemodynamic responses to PEEP or recruitment maneuvers. Although most clinicians choose PEEP levels according to indices of arterial oxygenation,[75] there is preliminary evidence that favors PEEP adjustments guided by esophageal manometry.[93] Notwithstanding uncertainty about "mediastinal artifacts" in recumbent patients with pleural effusions and "heavy lungs,"[94,95] the pressure in the mid to lower esophagus represents an estimate of local, if not global, lung surface (or pleural, Ppl) pressure. As such, when referenced to airway pressure (Pao, a surrogate of alveolar pressure in the absence of flow) the esophageal pressure (Pes) informs about lung stress (i.e., transpulmonary pressure, Ptp). Remarkably, several reports indicate that up to 50% of patients with injured lungs have a negative Ptp at end expiration (Pes >Pao) despite PEEP settings as high as 20 cm H_2O (Fig. 5-5).[82,93] It would be easy to dismiss these findings as artifact, were it not for a small prospective randomized clinical trial that hinted at a survival benefit in patients in whom PEEP therapy was targeted to an end-expiratory Ptp of +5 cm H_2O.[93] It suggests that in the recumbent posture the injured lung is mass loaded by the chest wall and resists emptying on account of interfascial forces from alveolar or airway fluid and foam.[56] It may also mean that the risks of overinflating nondependent lung units, which are undoubtedly stressed by the more aggressive PEEP approach, is not as great as generally thought.

Figure 5-5



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Relationship between airway pressure (*x axis*) and esophageal pressure (*y axis*) at end-expiration (*solid circles*) and end-inspiration (*open triangles*) in mechanically ventilated patients with ALI. As a group, most patients were managed at PEEP settings ≥10 cm H_2O, yet the corresponding esophageal pressure exceeded PEEP in 50% of instances. This suggests that the lungs were compressed by the chest wall and not sufficiently recruited at end-expiration. (Used, with permission, from Talmor,et al. Esophageal and transpulmonary pressures in acute respiratory failure. *Crit Care Med.* 2006;34:1389.)

For those who rely on acute changes in Pa_{O_2} as surrogate end points of PEEP management, certain caveats are in order. In critically ill patients with injured lungs, Pa_{O_2} is sensitive to changes in metabolic rate and cardiac output.[96,97] Because patients with injured lungs have $\dot{V}/\dot{Q}$ mismatch as well as shunt, changes in mixed venous oxygen tension, which result from changes in metabolic rate and cardiac output, must influence arterial P_{O_2}. Therefore, the P_{O_2} response to step changes in PEEP (and/or recruitment maneuvers) is determined by a net balance between positive and negative effects. Positive effects include (a) reductions in the number of lung units with low $\dot{V}/\dot{Q}$ and shunt as a result of their recruitment, (b) increases in cardiac output driven by the sympathomimetic effects of CO_2 retention (the latter invariably accompanies recruitment maneuvers), and (c) a fall in oxygen uptake associated with respiratory acidosis.[98,99] Negative effects include (a) increases in low $\dot{V}/\dot{Q}$ and shunt in patients who are PEEP-resistant and in whom the increased alveolar pressure diverts blood away from normal lung toward diseased lung,[100] (b) a fall in cardiac output resulting from volume- and pressure-mediated decreases in venous return,[101] (c) a fall in cardiac output resulting from volume-mediated and pressure-mediated increases in pulmonary artery pressure and right-ventricular afterload,[102] and (d) increases in systemic oxygen consumption as a behavioral response to increased lung expansion and CO_2 retention. The cardiovascular and metabolic confounders of the recruitment response may be deduced from pulse and blood-pressure responses. Alternatively, lung recruitment ought to result in a change in respiratory system mechanics. The clinician, however, should be under no illusion that such change will be large and easy to discern from peak and plateau pressure measurements.[79] This is so because comparisons between states require careful attention to muscle relaxation and the matching of volume and time histories.[103] Finally, because clinicians generally must rely on pulse oximetry as opposed to online P_{O_2} measurements, they must consider the time delays secondary to circulation time and signal processing when assessing the recruitment response.[104]

Choosing the Appropriate Tidal Volume

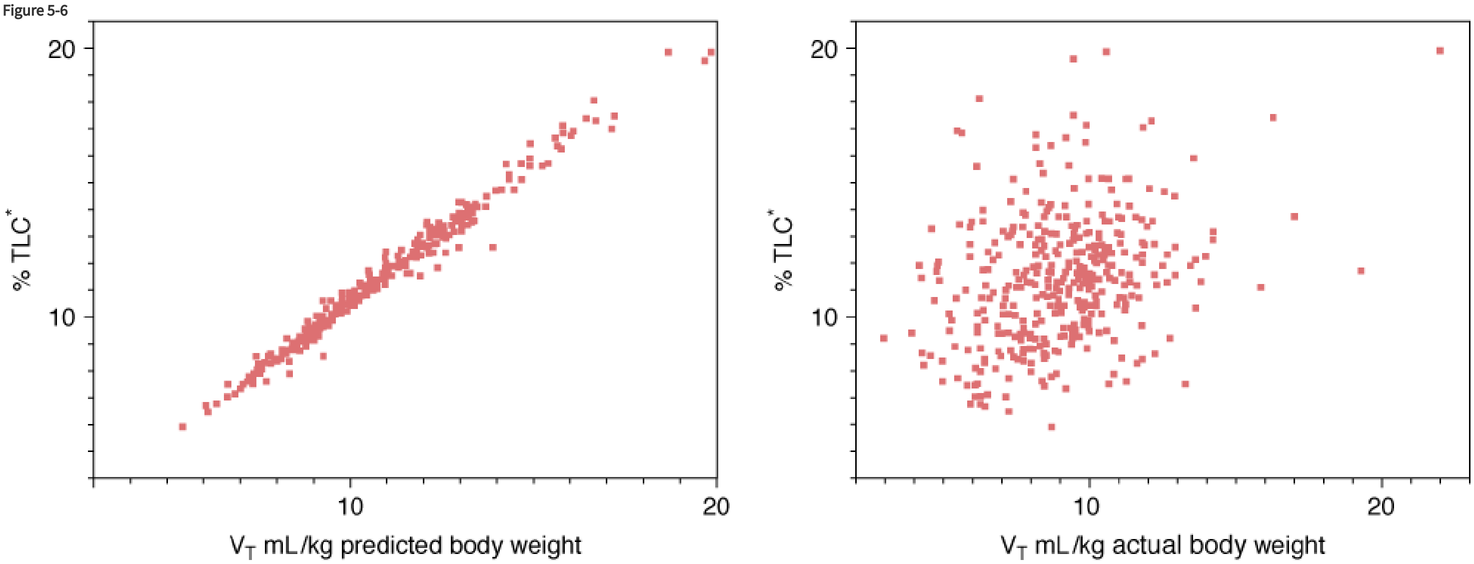
The choice of V_T is arguably the most important decision a care provider makes when initiating mechanical ventilation. For many years, physicians have chosen ventilator V_T between 10 and 15 mL/kg of actual body weight. This recommendation can be traced back to the early days of positive-pressure ventilation, when this therapy was reserved for patients with neuromuscular diseases, such as poliomyelitis. Patients with near-normal lungs feel more comfortable when they are ventilated with two to three times normal V_T. In patients with injured lungs, however, a V_T of as little as 10 mL/kg of actual body weight can have devastating effects on lung structure, function, and, ultimately, outcome.[33,34,50,105]

To fully appreciate the importance of V_T settings, it is useful to consider distinct physical lung-injury mechanisms: (a) regional overinflation, caused by the application of a local stress or pressure that forces cells and tissues to assume shapes and dimensions that exceed those experienced during even the most strenuous exercise,[106,107] (b) so-called low-volume injury associated with the repeated opening and closure of unstable lung units,[57–59] (c) inactivation of surfactant, on account of large alveolar surface-area oscillations,[108,109] and (d) interdependence mechanisms that raise cell and tissue shear stress between neighboring structures with differing mechanical properties.[110]

The injured lung is particularly susceptible to physical damage because its inspiratory capacity is reduced and its dorsal units tend to get obstructed with liquid plugs.[111,112] As a result, the greater the V_T, the greater is the likelihood that the lung will be damaged by both high-volume and low-volume injury mechanisms. One approach that requires no judgment whatsoever is simply to adopt in all patients with injured lungs the settings of the low V_T arm of the ARDS Network clinical trial,[34] which established the efficacy of lung-protective mechanical ventilation. Patients randomized to the low V_T arm received a V_T of 6 mL/kg of predicted body weight. If their end-inflation pause pressure exceeded 30 cm H_2O, then V_T was reduced further to as little as 4 mL/kg of predicted body weight. In patients in whom 6 mL/kg of predicted body weight resulted in breath stacking, effectively doubling their V_T, and in whom stacking could not be abolished with sedation, V_T was increased up to 8 mL/kg of predicted body weight. This approach was associated with a 23% reduction in all-cause mortality compared with a high-V_T strategy.[34]

The uniform adoption of the ARDS Network recommendation to set V_T to 6 mL/kg predicted body weight in all patients with injured lungs has been challenged.[85,113] Epidemiologic studies have established height and gender as opposed to actual body weight as the best predictors of absolute lung volume, including total lung capacity (TLC).[114,115] The ARDS Network investigators predicted ideal body weight from an equation based on height and gender. A graphic comparison of the two predictive equations (Fig. 5-6) shows that 1 mL/kg of predicted body weight corresponds to 1% predicted TLC. Therefore, the recommendation to restrict V_T of patients with injured lungs to 6 mL/kg of predicted body weight amounts to restricting V_T during mechanical ventilation to no more than 6% of preinjury TLC. The right-hand side of the figure shows that there is no correlation between predicted TLC and actual body weight in a population of patients who were ventilated at the Mayo Clinic in 2001.[105] Although these observations underscore the fallacy of scaling V_T to actual body weight, one may reasonably argue that scaling V_T to predicted body weight also misses the mark. To the extent to which the treatment objective of lung-protective mechanical ventilation is to minimize lung stretch, one would want to scale V_T to the capacity of the injured lung and not that of the lung before it was injured. It is well established that the injured lung has fewer recruitable lung units than a normal lung, hence the analogy to "baby lung," a term coined by Gattinoni.[112] Given the variability in lung impairment and hence lung capacity between patients with ALI, it is not surprising that a seemingly uniform V_T setting of 6 mL/kg predicted body weight produces very different parenchymal deformations in a population with lung disease.[85,113]

Figure 5-6



Predicted or ideal (*left panel*) and actual body weights (*right panel*) of 332 mechanically ventilated patients plotted against their predicted normal total lung capacity (%TLC). The predictive equations for ideal body weight and TLC are based on height and gender. Not surprisingly, the correlation between predicted body weight and predicted TLC is excellent. Note, however, that the correlation between actual body weight and percent TLC is extremely poor. The source data are from patients included in a report by Gajic et al.[105]

Lung tissue deformation can be quantified as strain. A strain is a normalized measure of deformation representing the displacement between particles in the body relative to a reference length. A recent report suggests that in normal anesthetized and mechanically ventilated pigs, lung damage occurs only when a strain greater than 1.5 to 2.0 is reached or overcome, implying that normal lungs are quite resistant to ventilator-induced injury.[116] Strain was defined as the fractional volume change between functional residual capacity and the thoracic gas volume at end-inflation. Because none of the animals were ventilated with PEEP, strain equaled V_T/functional residual capacity. Although these data are reassuring for anesthesia practice, it is important to remember that the threshold for strain injury of 2.0 may not hold in instances in which the provider increases end-expiratory lung volume through the use of PEEP. Be this as it may, it must be understood that strain is critically dependent on the choice of reference volume.

Even though there is a powerful rationale for scaling V_T to the capacity of the injured lung, lung capacity or the number of recruitable lung units is rarely measured in current practice. The recommendation to limit airway inflation pressure to ≤30 cm H_2O reflects concerns about overstretching the lung,[117] but ignores the highly variable influence of the chest wall on lung volume and respiratory system mechanics in ALI.[118] Lung capacity, that is, the size of the “baby lung,” can be inferred from measurements of thoracic gas volume at a defined airway pressure using either gas-dilution methods or computed tomography.[113,119] Alternatively, the inspiratory capacity can be inferred from the volume of gas that enters the lung during an inflation from 5 to 40 cm H_2O.[84] Although this approach is certainly feasible, there is currently no consensus as to what fraction of capacity V_T may safely occupy.

The results of the low-volume ventilation ARDS Network trial generated a heated debate as to whether outcome differences reflected the obsolete management of the high V_T group or improved management of the low V_T group.[120] Be this as it may, the debate produced some important questions: (a) Is the choice of V_T also important in patients who are ventilated with what generally are considered “safe” inflation pressures? (b) Are ventilator modes in which diaphragmatic activity is preserved superior to the low V_T approach used in the ARDS Network trial? (c) Should one care about V_T restrictions in patients with lung diseases other than ALI?

1. *Is the choice of V_T also important in patients who are ventilated with what generally are considered “safe” inflation pressures?* Inflating the lungs beyond TLC greatly increases the risk for barotrauma. Barotrauma is characterized by extraalveolar air and is an entity distinct from ventilator-induced lung injury.[50,51] In upright man, the transpulmonary and alveolar pressures at TLC approximate 25 and 35 cm H_2O, respectively.[121] The lungs of recumbent patients with ALI, however, are frequently not fully inflated at these pressures.[122] There is an ongoing debate as to whether mechanical ventilation with plateau pressures of less than 30 cm H_2O is safe irrespective of the choice of V_T. None of the available clinical and experimental studies is sufficiently convincing to base general management recommendations on. In the absence of convincing data, one should exercise extreme caution when departing from low V_T guidelines in patients with ALI. Indeed, circumstantial evidence and reasoning favor strict adherence to low V_T guidelines in patients with ALI because (a) spontaneous hyperventilation, which by definition never exceeds TLC, has been implicated as a cause of surfactant dysfunction and noncardiogenic pulmonary edema,[123] (b) repeated inflations of the respiratory system to pressures and volumes below TLC but above the upper inflection point of the inflation P-V curve are associated with lung injury,[81,116] and (c) a post hoc analysis of ARDS Network data suggested that patients in all plateau pressure quartiles derived benefit from V_T reductions.[124]

2. *Are ventilator modes in which diaphragmatic activity is preserved superior to the low V_T approach used in the ARDS Network trial?* Patient-assisted breathing modes, such as PSV, bilevel pressure ventilation, and APRV or NAVA, have been touted as modes of choice in the management of patients with ALI.[125,126] The evidence in support of this recommendation is not as strong as that in support of the low V_T strategy employed in the ARDS Network trial.[34] The rationale for partial support centers on the increased regional ventilation and reduced incidence of atelectasis in dorsal lung regions when diaphragmatic activity is preserved.[127,128] Whether this observation has bearing on patient survival, however, is currently unclear. It should be noted that, on general principles, V_T-related effects on lung structure and function, including injury mechanisms, are not specific to ventilator mode. Until proven otherwise it should be assumed that a V_T of 12 mL/kg predicted body weight is potentially injurious to the lungs regardless whether the patient breathes spontaneously, is mechanically ventilated in a low-pressure preset support mode, or is paralyzed and fully supported in a volume-controlled mode.

3. *Should V_T be restricted in patients with respiratory failure from conditions other than ALI?* To the extent to which lung strain and alveolar overdistension are the prevailing injury mechanisms, lungs with relatively preserved inspiratory capacity are much less susceptible to deformation injury.[116] To date, several prospective clinical trials in search of associations between intraoperative ventilator settings and biomarkers of lung stress have either uncovered no significant association or favor a low V_T strategy.[129,130] A retrospective cohort study,

however, of patients who were mechanically ventilated for more than 48 hours and who did not have ALI from the outset, identified V_T as a major risk factor for the subsequent development of noncardiogenic pulmonary edema.[105] This association was recently confirmed in a prospective clinical trial.[131] Because there is no compelling reason why any patient with normal or near-normal lungs would benefit from or need a V_T of greater than 10 mL/kg of predicted body weight, V_T settings above this threshold should be used with caution.

Respiratory Rate

Having settled on a V_T and an end-expiratory volume, adjustments in the machine backup rate (f_M) should be made considering: (a) the patient's actual rate demand, (b) the patient's anticipated ventilatory requirement, and (c) the impact of the rate setting on breath timing (see Fig. 5-1). Virtually all patients with hypoxic respiratory failure are tachypneic and usually require f_M settings of between 20 and 30 breaths per minute. Unless the patient has been paralyzed or has been so heavily sedated that spontaneous inspiratory triggering efforts are not sensed, f_M settings of 20 breaths per minute or lower are poorly tolerated because: (a) neurohumoral feedback from lung edema and inflammation induces rapid shallow breathing independent of chemoreceptive and mechanoreceptive effects on central pattern generation, (b) in the presence of a severe gas-exchange impairment, low rates and minute volumes would cause CO_2 retention, which, in turn, elicits its own disease state–independent effects on respiratory rate and drive, and (c) discrepancies between actual (triggered) and set machine (backup) rate promote breathing patterns with inverse inspiratory-to-expiratory timing ratios and double triggering. Inverse breathing-pattern ratios are not compatible with normal phase-switching mechanisms in the presence of respiratory distress. Conventional ventilator modes are not capable of varying T_I or inspiratory flow with the actual machine rate. For example, at an f_M setting of 10 breaths per minute, the T_{TOT} is 6 seconds. If the I:E ratio is set at 1:2, or if V_T and flow have been set to 0.5 and 0.25 L/s, respectively, then T_I is fixed at 2 seconds, and expiratory time (T_E) will be 4 seconds. If the patient actually triggers at 20 breaths per minute, then T_{TOT} declines to 3 seconds. T_I remains fixed at 2 seconds because it is determined by the preset machine (backup) rate, the I:E ratio, or the inspiratory-flow setting. T_E now must decrease from 4 seconds to 1 second, and the actual I:E ratio will increase from 1:2 to 2:1. At a rate of 30 breaths per minute (T_{TOT} = 2 seconds), T_E becomes 0, and "fighting the ventilator" must result. For these reasons, the f_M always should be set close to the patient's actual rate. If the actual rate is so high that effective ventilation cannot be achieved, then the patient needs additional sedation and possibly neuromuscular blockade. However, it should be emphasized that some patients, while ventilated in closed-loop modes, such as PAV or NAVA, chose rates in the high 30s and 40s, but maintain adequate gas exchange and sustainable workloads. Under these conditions, tachypnea need not imply discomfort or inadequate ventilator support!

Timing Variables

I:E Ratio

The setting of timing variables in conjunction with V_T and extrinsic PEEP determines the volume range over which lungs are cycled during ventilation. A long T_I, a high T_I/T_{TOT}, and a low mean inspiratory flow all promote ventilation with an inverse I:E ratio. Despite the considerable number of endorsements of inverse-ratio ventilation in ARDS, the beneficial effects of increasing I:E beyond 1:1 on pulmonary gas exchange tends to be marginal, provided that V_T and end-expiratory volumes are held constant.[132] All ventilators provide the option of maintaining lung volume at end inflation through the use of an inspiratory-hold time or pause time that usually is expressed as a percentage of the total cycle time (%T_{TOT}). For the purpose of defining the I:E ratio, the pause time is considered part of the inspiratory machine cycle. Long pause times favor the recruitment of previously collapsed or flooded alveoli and offer a means of shortening expiration independent of rate and mean inspiratory flow ($\dot{V}_1$). Although alveolar recruitment is a desired therapeutic end point in the treatment of patients with edematous lungs, one should at least consider that keeping the lungs expanded at high volumes (and pressures) for some time may damage relatively normal units and may cause adverse hemodynamic effects.

Inspiratory Flow

Most ventilators require that mean $\dot{V}_1$ and its profile be specified. Mean $\dot{V}_1$ is equal to the ratio of V_T to T_I. Therefore, one cannot change flow without affecting at least one of the other timing variables (see Fig. 5-1). It is also important to consider that changing the flow profile from a square wave to a decelerating or sine-wave pattern prolongs T_I in ventilators that require a peak-flow setting. This is so because non–square-wave profiles have a higher peak-to-mean flow ratio; that is, it takes longer to deliver the predefined V_T than in square-wave flow delivery modes. Unless the patient is struggling, mean $\dot{V}_1$ usually is set to no more than 1 L/s during volume-preset ventilation. In patients in whom lung recruitment and oxygenation are the primary therapeutic end points, setting flow (and rate) so that the T_I/T_{TOT} approximates 0.5 (I:E = 1) tends to achieve the goal. Increasing flow always will raise peak airway pressure, but this need not be of concern if most of the added pressure is dissipated across the endotracheal tube. On the other hand, there is experimental evidence that the rate of lung expansion is a V_T-independent risk factor for lung deformation injury. Although

$\dot{V}_I$ is one of the factors that determine the regional distribution of inspired gas,[41] the lung volume–independent effects of flow on pulmonary gas exchange are too unpredictable to warrant general guidelines. Much more important is the realization that the combined effects of flow, volume, and time settings influence the functional residual capacity and the degree of dynamic hyperinflation.[40,133,134]

Mean Expiratory Flow: The Hidden Variable

Mean expiratory flow is defined by the ratio of V_T to T_E. $T_E = T_{TOT} - T_I$. $T_{TOT} = 60/f$ (per minute). Because the f_M and the actual f may differ from each other in the assist-control mode (AC), the assumed and the actual T_{TOT} also may differ. Recall from the discussion on rate and timing that T_I is defined by both the set f_M and the set I:E ratio and that T_I remains constant irrespective of the actual rate. In contrast, T_E is affected by the actual breathing rate (f_A): $T_E = 60/f_A - T_I$. Therefore, the choice of volume and timing settings, together with the patient's trigger rate, determine mean expiratory flow. V_T/T_E is the principal ventilator setting-related determinant of dynamic hyperinflation. A patient with airway obstruction and a maximal forced expiratory flow (FEF) of 0.2 L/s in the mid–vital capacity range, and obviously cannot accommodate a V_T/T_E of 0.5 L/s without an increase in end-expiratory lung volume (see "Obstructive Lung Diseases" below).

Minute Ventilation

In general, minute ventilation ($\dot{V}_E$) is not a variable that is set directly by the operator, but it is the consequence of the V_T and rate settings. $\dot{V}_E$ is an important determinant of the body's CO_2 stores and consequently of the arterial CO_2 tension (Pa_{CO_2}):

$$Pa_{CO_2} = \frac{\dot{V}_{CO_2} \times k}{\dot{V}_E(1 - V_D/V_T)} \qquad (8)$$

where $\dot{V}_{CO_2}$ = volume of CO_2 produced in liters per minute; V_D/V_T = dead-space-to-tidal-volume ratio, a variable with which the efficiency of the lung as a CO_2 eliminator can be approximated; and k = 0.863 and is a constant that scales $\dot{V}_{CO_2}$ and $\dot{V}_E$ to the same temperature and humidity.

If the main goal of mechanical ventilation were to normalize Pa_{CO_2}, then $\dot{V}_E$ would be the most important machine setting. Although a "normal" Pa_{CO_2} is one of the therapeutic end points of mechanical ventilation, at times, normocapnia can be achieved only with high lung inflation volumes and pressures. This is particularly true in patients with ALI because they are often hypermetabolic (high $\dot{V}_{CO_2}$) and in addition suffer from $\dot{V}/\dot{Q}$ mismatch (high V_D/V_T).[135] For these reasons, it is not unusual to encounter patients with ALI whose $\dot{V}_E$ requirements exceed 20 L. In the past, concerns about acid–base status dominated the choice of ventilator settings. In recent years, however, the focus on mechanical lung injury has resulted in a reappraisal of therapeutic priorities that now places the prevention of lung injury above the goal to normalize CO_2 tensions and acid–base status. The corresponding ventilation strategy has been termed *permissive hypercapnia.*[136,137] Permissive hypercapnia means that the physician accepts a Pa_{CO_2} outside the expected or "normal" range in order to minimize the potential for ventilator-induced lung injury. Because such a ventilation strategy runs contrary to the limits set by the chemoresponses of neural ventilatory control mechanisms, permissive hypercapnia usually requires heavy sedation, and sometimes paralysis of the patient. Until recently, most providers considered neuromuscular blockade as an intervention of last resort; several clinical trials, however, by the same team suggest a survival benefit associated with early, time-limited neuromuscular blockade in patients with severe ARDS.[138]

There is a great deal of interest in the consequences of hypercapnia on pulmonary vascular barrier function, signaling mediated by reactive oxygen and nitrogen species, innate immunity, and ultimately, patient survival.[2–7,139–141] The science is fascinating, but it is not sufficiently advanced to derive clinical management decisions. That said, most experts probably would agree that (a) there is no universal pH or P_{CO_2} threshold that mandates a corrective action, (b) the use of bicarbonate buffers to correct respiratory acidemia is unproven, and (c) tracheal gas insufflation generally is effective in reducing Pa_{CO_2} by 10 mm Hg or less.[142,143] Renewed interest in extracorporeal membrane oxygenation as an adjunct to lung-protective mechanical ventilation may well alter the current approach to permissive hypercapnia in years to come.[144] Except for several extracorporeal membrane oxygenation centers with high patient volumes, however, this intervention should be considered experimental at this point in time.[145]

Obstructive Lung Diseases

In patients with obstructive lung diseases, there is a reduced capacity for generating expiratory flow. When obstruction is severe enough to cause ventilatory failure, dynamic airway collapse is virtually always present during the expiratory phase of the ventilatory cycle.[10,146] This means that the passive elastic recoil forces of the relaxed respiratory system are large enough to produce maximal expiratory flows in the tidal breathing range. Such patients are prone to dynamic hyperinflation, which may adversely affect circulation,[147] may increase the risk of barotrauma,[148] and can place the diaphragm and inspiratory muscles at a mechanical disadvantage.[149–151] Consequently, the primary therapeutic goal of mechanical ventilation in obstructive lung disease is to minimize the thoracic volume about which the lungs are ventilated. Additional goals vary with the context in which airflow obstruction is observed. Patients with long-standing obstruction from emphysema or chronic bronchitis (unless they are "fighting the ventilator") usually are easy to ventilate and simply may need respiratory muscle rest and a resetting of the CO_2-response threshold to more normal values. These secondary therapeutic objectives are highly controversial. In contrast, patients with acute severe asthma often "fight the ventilator" and therefore often require sedation, neuromuscular blockade, and ventilation with permissive hypercapnia.[136,148] Such patients are prone to neuromuscular insults from glucocorticoids and paralytic agents and may require prolonged mechanical ventilation for weakness long after lung mechanics normalize.[152]

Minimizing Dynamic Hyperinflation

The ventilator management of patients who are prone to dynamic hyperinflation is best understood after a review of the expiratory mechanics of the relaxed respiratory system (see "The Mechanical Determinants of Patient–Ventilator Interactions" above). The key determinants of end-expiratory lung volume in a ventilated patient are the time constant of the respiratory system (R × C) and the V_T/T_E that has been imposed by the ventilator settings.[40] Figure 5-6 underscores these concepts, which are fundamental to formulating a meaningful management plan. If it is assumed that a mechanical inflation of 1 L is initiated from Vrel at a rate of 20 breaths per minute and an I:E ratio of 1:2, the patient has 2 seconds to exhale. In the example in Figure 5-6, the maximal mean passive expiratory flow that can be achieved in this volume range (between Vrel and Vrel + 1 L) is given by the expiratory flow-volume curve. In this example, the maximal mean flow is only 0.25 L/s. Hence, in the 2 seconds available for expiration, the patient can exhale only half the inspired volume (0.5 L) before the next inflation is initiated by the machine. Thus, the second breath is begun at a lung volume of Vrel + 0.5 L. Maximal mean expiratory flow over the new volume range (Vrel + 0.5 L and Vrel + 1.5 L) is 0.3 L/s. This flow is still insufficient for adequate lung emptying. A new steady state will be achieved only when the increase in lung volume results in a maximal mean expiratory flow of 0.5 L/s, which is equal to the obligatory mean expiratory flow imposed by the ventilator settings.

Dynamic hyperinflation is associated with an increase in alveolar pressure at end expiration. This pressure, also called *intrinsic positive end-expiratory pressure* (PEEPi), is the pressure of the respiratory system at end expiration plus any pressure generated by respiratory muscles.[10,153] In the absence of muscle activity, the degree of dynamic hyperinflation can be inferred from the end-expiratory airway occlusion pressure (PEEPi) and the elastance of the relaxed respiratory system (Ers):

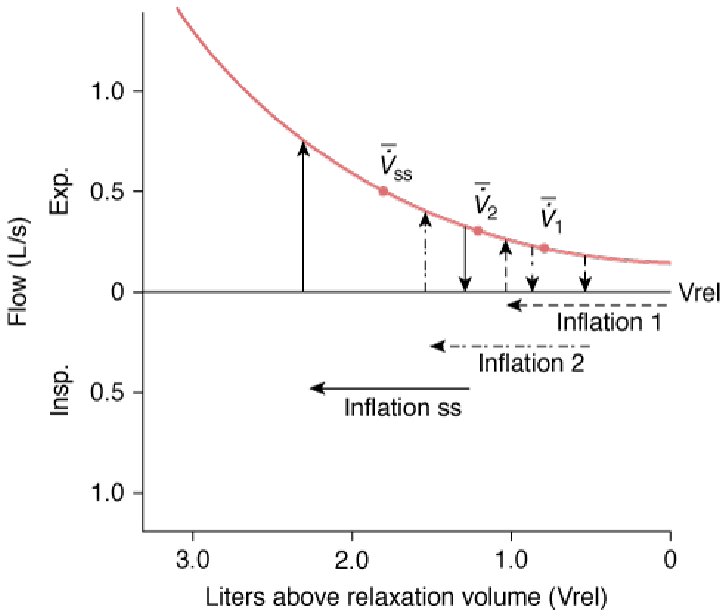
$$\mathrm{Vee} - \mathrm{Vrel} = \mathrm{Ers}/\mathrm{PEEPi} \qquad (9)$$

where Vee is the volume of the lungs at end expiration.

In the presence of muscle activity from active expiration or inspiratory triggering efforts, PEEPi is a meaningless measurement. This limitation also applies to some extent to esophageal pressure-derived estimates of PEEPi. In some ventilators, PEEPi can be estimated at the "press of a button"—by pressing an end-expiratory hold button and waiting until airway opening pressure reaches a steady value. In ventilators in which the timing of end-expiratory occlusions is not automated, the measurement of PEEPi is considerably more difficult.

As illustrated in Figure 5-7, ventilator adjustments designed to minimize dynamic hyperinflation should be geared toward lowering mean expiratory flow (V_T/T_E). In a paralyzed patient with asthma, V_T can be reduced to as little as about 4 mL/kg of predicted body weight, whereas TE is prolonged through increases in mean inspiratory flow (1 to 1.5 L/s), adjustments in the I:E ratio (1:4 to 1:5), and reductions in f_M (approximately 10 breaths per minute). As discussed earlier, such a strategy is likely to produce hypercapnia, but even severe acidemia is usually well tolerated in paralyzed subjects.[154] High inspiratory-flow settings, which are required to prolong T_E, are bound to increase peak Paw and may raise concerns about barotrauma. It must be emphasized, however, that much of this added "resistive pressure" is dissipated along the endotracheal tube and proximal airways and that, on balance, increasing the rate of lung inflation seems less damaging than ventilating asthmatic lungs near TLC. Consistent with this hypothesis, the incidence of barotrauma can be reduced significantly in patients with status asthmaticus when ventilator settings are chosen to maintain peak lung volume within 1.4 L of Vrel.[148,155,156]

Figure 5-7



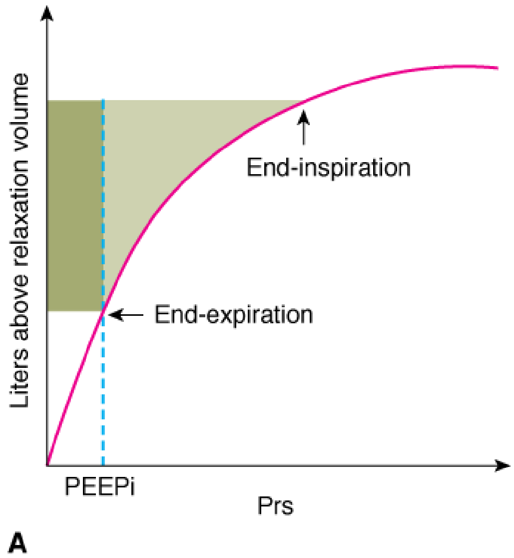
Diagrammatic demonstration of how insufficient expiratory flow produces dynamic hyperinflation. The *broken horizontal arrow* shows the first breath of 1 L initiated from relaxation volume (Vrel) at a rate of 20 breaths per minute and a T_I/T_{TOT}) of 0.33 (*inflation 1*). The *solid curved line* shows the maximal expiratory flow that can be produced during passive exhalation by the elastic recoil pressure of the system. In the 2 seconds available from expiration, the maximum mean expiratory flow of the first breath ($\bar{\dot{V}}_1$) is only 0.25 L/s. Therefore, only 0.5 L can be exhaled in the 2 seconds before the next inhalation of 1 L is initiated (*inflation 2*). According to the flow–volume relationship, a maximal mean expiratory flow of 0.3 L/s ($\bar{\dot{V}}_1$) can be achieved over this volume range. A steady state, during which inspiratory and expiratory volumes are matched, will be reached only when the maximal mean expiratory flow ($\bar{\dot{V}}_{ss}$) equals 0.5 L/s. (Used, with permission, from Hubmayr, et al. Physiologic approach to mechanical ventilation. *Crit Care Med.* 1990;18:103–113.)

Permissive hypercapnia and neuromuscular blockade are rarely required in patients with ventilatory failure from exacerbations of chronic obstructive lung diseases. Nevertheless, many such patients have respiratory rates in the high teens and low twenties, making it difficult to prolong T_E beyond approximately 2 seconds. This makes it virtually impossible to ventilate these patients near Vrel. Recall that patients with end-stage obstruction may have maximal expiratory flows of 0.2 L/s or less up to volumes near TLC.[146] In the nonparalyzed patient, hypercapnia sets limits to the reductions in V_T; consequently, attempts must be made to reduce the patient's respiratory rate. Sometimes the only way to minimize hyperinflation without having to resort to neuromuscular blockade is through the judicious use of sedatives with the intent of reducing inspiratory efforts until they fail to initiate a machine breath (see "Asynchrony Between the Patient's Effort and Machine-Delivered Breaths" below).

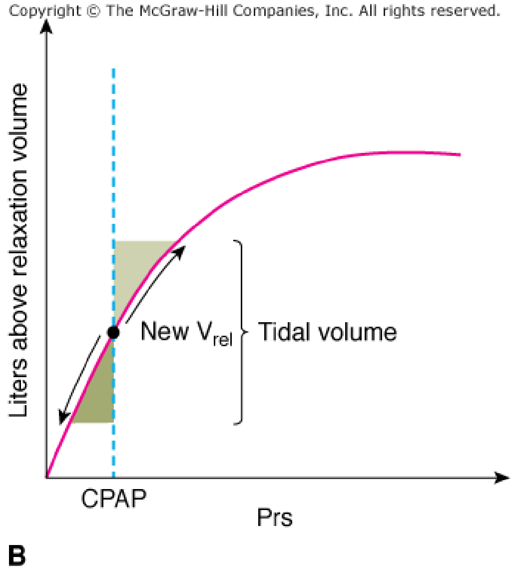
Use of Continuous Positive Airway Pressure

In patients with hypoxic respiratory failure, continuous positive airway pressure (CPAP) is used to raise lung volume to recruit closed and flooded alveoli and to improve oxygenation. In contrast, the goal of CPAP therapy in patients with obstruction is to minimize inspiratory work.[157] Figure 5-8A shows the potential mechanisms of action of CPAP in obstructed patients schematically. The figure shows the pressure–volume relationships of the relaxed respiratory system and depicts the elastic work (Wel) needed to raise lung volume from end expiration to end inspiration (shaded area) in the presence of dynamic hyperinflation. Wel has two components: (a) work required to halt expiratory flow by counterbalancing respiratory system recoil at end expiration (W related to PEEPi) and (b) work expended during inflation of the lungs and thorax. In theory, the inspiratory work related to PEEPi (darker shaded area) can be provided externally with CPAP equal to PEEPi. As CPAP approaches PEEPi, however, additional hyperinflation may occur.[158] To guard against CPAP-induced worsening of hyperinflation, the physician can monitor peak or end-inflation hold pressure as an indicator of peak lung volume.

Figure 5-8



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

A. Effect of dynamic hyperinflation of elastic inspiratory work. The *solid curve* shows the relationship between the volume above Vrel and the recoil of the respiratory system (Prs). Dynamic hyperinflation exists. Inspiration is now initiated from a volume above Vrel. The increase in lung volume necessitates an increase in the elastic inspiratory work, which may be considered to have two components: work to halt expiratory flow (*darker-shaded area*) and work required to inflate the respiratory system (*lighter-shaded area*). **B.** Effect of CPAP on respiratory work. The *solid curve* is the pressure–volume curve of the respiratory system. With CPAP, a new Vrel is achieved. To conserve inspiratory elastic work, the patient recruits expiratory muscles and exhales below the new Vrel. The elastic work performed by the expiratory muscles is represented by the *darker-shaded area.* Relaxation of the expiratory muscles inflates the lungs back to the new Vrel without inspiratory effort. The inspiratory muscles then increase lung volume further, performing elastic inspiratory work *(lighter-shaded area*). Hence CPAP reduced the work of the inspiratory muscles by letting the expiratory muscles do part of the inspiratory work. (Used, with permission, from Hubmayr, et al. Physiologic approach to mechanical ventilation. *Crit Care Med.* 1990;18:103–113.)

Figure 5-8B shows an alternative mechanism by which CPAP may reduce inspiratory Wel. CPAP may result in exhalation below the new Vrel through the recruitment of expiratory muscles.[153,159] Subsequent relaxation of the expiratory muscle inflates the lungs passively back to the new Vrel. Inspiratory muscles are unloaded because the expiratory muscles do part of the inspiratory work. This is depicted by the lighter-shaded area in Figure 5-7B. This mechanism is of limited value in patients with severe obstruction, however, because low maximal flows prevent significant reductions in lung volume below Vrel.

Ventilatory Pump Failure and Chronic CO_2 Retention

Resting the Respiratory Muscles

In the 1970s and 1980s, much emphasis was placed on respiratory muscle fatigue as a common cause of ventilatory failure.[160] Experimental evidence that this truly occurs in a clinical setting remains elusive.[161] Without addressing all the pros and cons of minimizing the patient's contribution to inspiratory work, evidence is mounting that mechanical ventilation inhibits respiratory motor output primarily through mechanoreceptive pathways. Studies on volunteers and patients have shown that depending on state and ventilator settings, spontaneous respiratory muscle activity can be abolished, reduced, or entrained to the ventilator.[162–164] Two respiratory control aspects of patient–ventilator interactions deserve particular emphasis. First, volume-preset mechanical ventilation at settings that normalize blood-gas tensions provides no safeguard against excessive respiratory work.[44] This means that ventilating patients in a volume-preset assist-control mode offers no universal guarantee for sufficient respiratory muscle rest. Second, sleeping and obtunded patients are susceptible to PSV setting-induced central apneas.[26–29] This can lead to problems if a clinician feels compelled to increase ventilator support without a mandatory backup in order to reduce the work of breathing at night. If this is done through low IMV backup rates, then apneas may trigger ventilator alarms and cause arousal and sleep fragmentation.[25]

Resetting the Chemostat

It remains controversial whether patients with chronic hypercapnia and complicating acute ventilatory pump failure should be "mechanically" hyperventilated to normocapnia in an attempt to restore normal chemoresponsiveness. Proponents of such a ventilator strategy may argue that CO_2 has negative inotropic effects on respiratory muscles[165] and that experience with nocturnal ventilator assistance suggests that resetting CO_2 responsiveness is feasible in some instances.[166] Opponents argue that maintenance of normocapnia in the presence of lung disease requires a high minute volume, which could represent a fatiguing load on the respiratory muscles. As a general rule, at the time weaning is contemplated, sustainable CO_2 tensions should range between 40 and 50 mm Hg. Patients who are being weaned with CO_2 tensions in the sixties tend not to do well and are severely limited.

Approaches to Common Potentially Adverse Patient–Ventilator Interactions

Respiratory Alkalosis

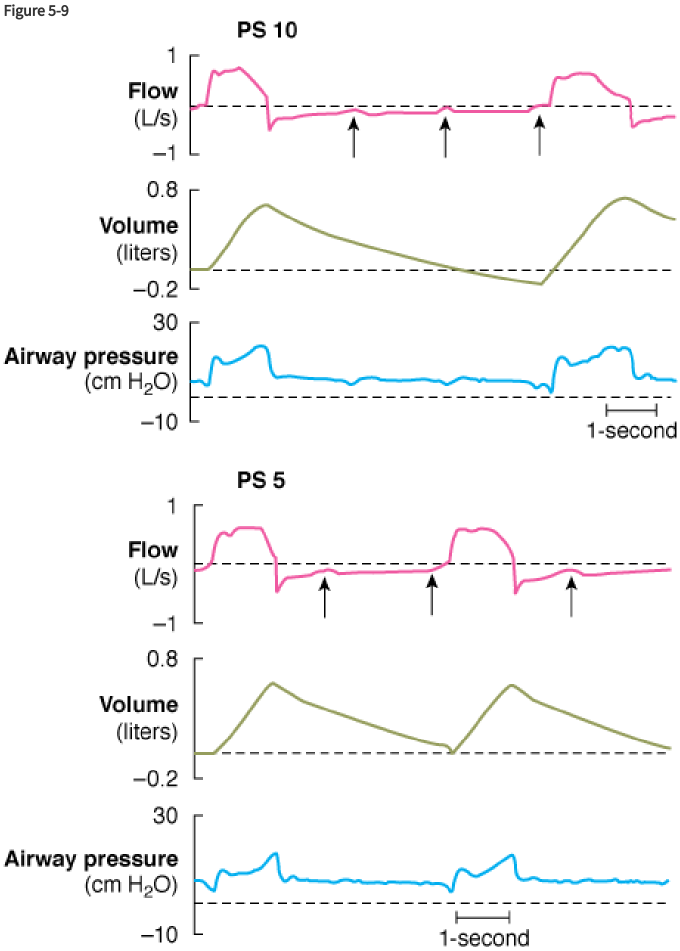
In spontaneously breathing normal subjects, $\dot{V}_E$ is closely coupled to Pa_{CO_2} and reflects both rate and V_T responses of the ventilatory control system. In mechanically ventilated subjects, V_T is often preset, thereby uncoupling ventilation from respiratory drive and confining the influence of neural control on the regulation of breathing to machine trigger rate. Consequently, ventilated patients with high intrinsic respiratory rates can have CO_2 tensions significantly below normal. Because associated alkalemia may contribute to arrhythmias and cardiovascular instability,[167] patients often are sedated, and the ventilator settings are adjusted with the goal of raising Pa_{CO_2}. In most instances of ventilator-induced hypocapnia, tachypnea is unrelated to hypoxemia or increased CO_2 drive per se. The causes of tachypnea may be behavioral in origin, as with pain and anxiety syndromes, or neurohumoral in origin, as with circulatory failure or in conjunction with lung and airway inflammation. Because tachypnea and increased ventilatory drive rarely are caused by CO_2 itself, any means of reducing ventilator-delivered volumes is effective in raising Pa_{CO_2}.

Asynchrony between the Patient's Effort and Machine-Delivered Breaths

Asynchrony between vigorous spontaneous efforts and machine-delivered breaths is often referred to as "fighting the ventilator." Because inspiratory efforts often are followed by active expiration in patients with increased drive, discrepancies between machine and patient T_I cause peak Paw to exceed the alarm (safety) limit (usually set to 45 cm H_2O), resulting in premature termination of inspiratory flow and insufficient ventilation. Although the initial management should be to raise the f_M and increase inspiratory flow up to 1.5 L/s, many patients with asynchrony require sedation and, on rare occasions, neuromuscular blockade. It is of note that the mechanisms of action by which sedatives facilitate mechanical ventilation have not been fully detailed. When "fighting the ventilator" reflects pain and anxiety, their mode of action is easily understood. Not all manifestations of respiratory distress, however, are behavioral in origin. It is not known to what extent various sedatives reduce the magnitude of inspiratory efforts (drive), slow respiratory rate, or facilitate the entrainment of medullary inspiratory pattern generation to the ventilator.

Asynchrony between patient and machine breaths is very common. This is particularly true for patients with high intrinsic respiratory rates, for patients with reduced inspiratory pressure output from low drive or respiratory muscle weakness, for patients with airways obstruction, and when ventilator support results in greater than normal V_T.[13,14,16,153] For example, Figure 5-9 shows pressure and flow tracings of a patient with airways obstruction and hypercapnic ventilatory failure during PSV of 10 and 5 cm H_2O. Arterial O_2 and CO_2 tensions were normal at both settings, and the patient did not appear to be in distress. The small deflections in expiratory flow marked by arrows represent inspiratory efforts (I) during the expiratory phase of the machine cycle. In the presence of dynamic hyperinflation, Pmus must counterbalance the expiratory recoil forces (Pel) before a new machine breath can be triggered. If ΔPmus is less than Pel minus the machine trigger sensitivity, then the inspiratory effort is wasted and does not result in a machine breath. At the PSV setting of 10 cm H_2O in this example, only every third inspiratory effort results in a machine breath (3:1 coupling). The low ΔPmus, the persistence of machine inflation after the cessation of inspiratory effort, and the presence of airways obstruction, with its propensity for dynamic hyperinflation, all contribute to machine trigger failure. Note that the reduction in PSV from 10 cm H_2O to 5 cm H_2O and the lower peak volume account for the reduced number of wasted inspiratory efforts. An awareness of this problem is important because the physician otherwise may attribute an increase in machine rate following reductions in PSV to impending failure or a fatiguing load response.

Figure 5-9



Flow, volume, and pressure tracings of a patient recorded during PSV with 10 cm H_2O (*upper panel*) and 5 cm H_2O (*lower panel*). Each *arrow* indicates an inspiratory effort. See the text for a further explanation. (Used, with permission, from Hubmayr RD. Coordinación de la musculatura respiratoria durante la desconexión de la ventilación mecánica en pacientes con enfermedades neurológicas (Respiratory muscle coordination during the weaning of patients with neurological diseases). In: Net A, Mancebo J, Benito S, eds. *Retirada de la ventilación mecánica.* Barcelona: Springer-Verlag Ibérica; 1995:164–181. With kind permission of Springer Science and Business Media.)

Because asynchrony between the patient and machine is common, its diagnostic and prognostic significance remains uncertain. When asynchrony impairs ventilator assistance or causes patient discomfort, treatment is required in the form of sedation and adjustments in CPAP, rate, flow, or trigger mode. When “wasted” inspiratory efforts are not perceived as uncomfortable, however, it is not clear that adjustments in ventilator settings are warranted. After all, increases in machine rate to match the rate of patient efforts may cause worsening dynamic hyperinflation and may compromise circulation. Alternatively, the use of large amounts of sedatives to assure synchrony cannot be considered harmless.[168] Patients who have been heavily sedated, but whose lung function is beginning to recover, can be particularly challenging insofar as their spontaneous tidal volumes often exceed “safe” levels. Consequently, imposing lung-protective V_T settings is commonly met with “double triggering,” in effect raising machine delivered V_T to 12 mL/kg predicted body weight. Because delirium and impaired airway protective reflexes often preclude extubation, the provider is faced with the difficult decision, whether to deepen sedation, institute neuromuscular blockade, reduce flow (i.e., prolong machine T_I) at the cost of flow-starving the patient and thereby increase the patient's work of breathing, or to ignore the high tidal volumes as a transient sedation/narcotics side effect. Although each of these options is defensible on theoretical grounds, most providers will slow deep breaths if liberation from endotracheal intubation is judged imminent.

References

1. Afshari A, Brok J, Moller AM, Wetterslev J. Inhaled nitric oxide for acute respiratory distress syndrome and acute lung injury in adults and children: a systematic review with meta-analysis and trial sequential analysis. *Anesth Analg.* 2011;112:1411–1421. [PubMed: 21372277]

2. Shibata K, Cregg N, Engelberts D, et al. Hypercapnic acidosis may attenuate acute lung injury by inhibition of endogenous xanthine oxidase. *Am J Respir Crit Care Med.* 1998;158:1578–1584. [PubMed: 9817711]

3. Laffey JG, Honan D, Hopkins N, et al. Hypercapnic acidosis attenuates endotoxin-induced acute lung injury. *Am J Respir Crit Care Med.* 2004;169:46–56. [PubMed: 12958048]

4. Lang JD, Figueroa M, Sanders KD, et al. Hypercapnia via reduced rate and tidal volume contributes to lipopolysaccharide-induced lung injury. *Am J Respir Crit Care Med.* 2005;171:147–157. [PubMed: 15477499]

5. Sinclair SE, Kregenow DA, Lamm WJ, et al. Hypercapnic acidosis is protective in an in vivo model of ventilator-induced lung injury. *Am J Respir Crit Care Med.* 2002;166:403–408. [PubMed: 12153979]

6. Vadasz I, Dada LA, Briva A, et al. AMP-activated protein kinase regulates CO2-induced alveolar epithelial dysfunction in rats and human cells by promoting Na,K-ATPase endocytosis. *J Clin Invest.* 2008;118:752–762. [PubMed: 18188452]

7. Wang N, Gates KL, Trejo H, et al. Elevated CO_2 selectively inhibits interleukin-6 and tumor necrosis factor expression and decreases phagocytosis in the macrophage. *FASEB J.* 2010;24:2178–2190. [PubMed: 20181940]

8. Jolliet P, Tassaux D, Roeseler J, et al. Helium-oxygen versus air-oxygen noninvasive pressure support in decompensated chronic obstructive disease: a prospective, multicenter study. *Crit Care Med.* 2003;31:878–884. [PubMed: 12627000]

9. Sassoon CS. Mechanical ventilator design and function: the trigger variable. *Respir Care.* 1992;37:1056–1069. [PubMed: 10145700]

10. Rossi A, Gottfried SB, Zocchi L, et al. Measurement of static compliance of the total respiratory system in patients with acute respiratory failure during mechanical ventilation. The effect of intrinsic positive end-expiratory pressure. *Am Rev Respir Dis.* 1985;131:672–677. [PubMed: 4003913]

11. Chatmongkolchart S, Williams P, Hess DR, Kacmarek RM. Evaluation of inspiratory rise time and inspiration termination criteria in new-generation mechanical ventilators: a lung model study. *Respir Care.* 2001;46:666–677. [PubMed: 11403698]

12. Murciano D, Boczkowski J, Lecocguic Y, et al. Tracheal occlusion pressure: a simple index to monitor respiratory muscle fatigue during acute respiratory failure in patients with chronic obstructive pulmonary disease. *Ann Intern Med.* 1988;108:800–805. [PubMed: 3369770]

13. Tobin MJ, Jubran A, Laghi F. Patient–ventilator interaction. *Am J Respir Crit Care Med.* 2001;163:1059–1063. [PubMed: 11316635]

14. Yamada Y, Du HL. Analysis of the mechanisms of expiratory asynchrony in pressure support ventilation: a mathematical approach. *J Appl Physiol.* 2000;88:2143–2150. [PubMed: 10846029]

15. Sassoon CS, Foster GT. Patient–ventilator asynchrony. *Curr Opin Crit Care.* 2001;7:28–33. [PubMed: 11373508]

16. Aslanian P, El Atrous S, Isabey D, et al. Effects of flow triggering on breathing effort during partial ventilatory support. *Am J Respir Crit Care Med.* 1998;157:135–143. [PubMed: 9445291]

17. Ishaaya AM, Nathan SD, Belman MJ. Work of breathing after extubation. *Chest.* 1995;107:204–209. [PubMed: 7813279]

18. Straus C, Louis B, Isabey D, et al. Contribution of the endotracheal tube and the upper airway to breathing workload. *Am J Respir Crit Care Med.* 1998;157:23–30. [PubMed: 9445274]

19. Esteban A, Frutos F, Tobin MJ, et al. A comparison of four methods of weaning patients from mechanical ventilation. Spanish Lung Failure Collaborative Group. *N Engl J Med.* 1995;332:345–350. [PubMed: 7823995]

20. Esteban A, Alia I, Gordo F, et al. Extubation outcome after spontaneous breathing trials with T-tube or pressure support ventilation. The Spanish Lung Failure Collaborative Group. *Am J Respir Crit Care Med.* 1997;156:459–465. [PubMed: 9279224]

21. Butler R, Keenan SP, Inman KJ, et al. Is there a preferred technique for weaning the difficult-to-wean patient? A systematic review of the literature. *Crit Care Med.* 1999;27:2331–2336. [PubMed: 10579244]

22. Bonmarchand G, Chevron V, Menard JF, et al. Effects of pressure ramp slope values on the work of breathing during pressure support ventilation in restrictive patients. *Crit Care Med.* 1999;27:715–722. [PubMed: 10321660]

23. Brochard L, Rauss A, Benito S, et al. Comparison of three methods of gradual withdrawal from ventilatory support during weaning from mechanical ventilation. *Am J Respir Crit Care Med.* 1994;150:896–903. [PubMed: 7921460]

24. MacIntyre NR, Cook DJ, Ely EW Jr, et al. Evidence-based guidelines for weaning and discontinuing ventilatory support: a collective task force facilitated by the American College of Chest Physicians; the American Association for Respiratory Care; and the American College of Critical Care Medicine. *Chest.* 2001;120:375S–395S.

25. Parthasarathy S, Tobin MJ. Effect of ventilator mode on sleep quality in critically ill patients. *Am J Respir Crit Care Med.* 2002;166:1423–1429. [PubMed: 12406837]

26. Morrell MJ, Shea SA, Adams L, Guz A. Effects of inspiratory support upon breathing in humans during wakefulness and sleep. *Respir Physiol.* 1993;93:57–70. [PubMed: 8367617]

27. Tobert DG, Simon PM, Stroetz RW, Hubmayr RD. The determinants of respiratory rate during mechanical ventilation. *Am J Respir Crit Care Med.* 1997;155:485–492. [PubMed: 9032183]

28. Branson RD, Johannigman JA, Campbell RS, Davis K Jr. Closed-loop mechanical ventilation. *Respir Care.* 2002;47:427–51; discussion 51–53. [PubMed: 11929616]

29. Brunner JX. Principles and history of closed-loop controlled ventilation. *Respir Care Clin N Am.* 2001;7:341–362, vii. [PubMed: 11517027]

30. Sinha SK, Donn SM. Volume-controlled ventilation. Variations on a theme. *Clin Perinatol.* 2001;28:547–560, vi. [PubMed: 11570154]

31. Branson RD, Johannigman JA. What is the evidence base for the newer ventilation modes? *Respir Care.* 2004;49:742–760. [PubMed: 15222907]

32. Iotti GA, Polito A, Belliato M, et al. Adaptive support ventilation versus conventional ventilation for total ventilatory support in acute respiratory failure. *Intensive Care Med.* 2010;36:1371–1379. [PubMed: 20502870]

33. Amato MB, Barbas CS, Medeiros DM, et al. Effect of a protective-ventilation strategy on mortality in the acute respiratory distress syndrome. *N Engl J Med.* 1998;338:347–354. [PubMed: 9449727]

34. Ventilation with lower tidal volumes as compared with traditional tidal volumes for acute lung injury and the acute respiratory distress syndrome. The Acute Respiratory Distress Syndrome Network. *N Engl J Med.* 2000;342:1301–1308.

35. Sinderby C, Navalesi P, Beck J, et al. Neural control of mechanical ventilation in respiratory failure. *Nat Med.* 1999;5:1433–1436. [PubMed: 10581089]

36. Kacmarek RM. Proportional assist ventilation and neurally adjusted ventilatory assist. *Respir Care.* 2011;56:140–148; discussion 149–152. [PubMed: 21333175]

37. Verbrugghe W, Jorens PG. Neurally adjusted ventilatory assist: a ventilation tool or a ventilation toy? *Respir Care.* 2011;56:327–335. [PubMed: 21255496]

38. Arold SP, Bartolak-Suki E, Suki B. Variable stretch pattern enhances surfactant secretion in alveolar type II cells in culture. *Am J Physiol Lung Cell Mol Physiol.* 2009;296:L574–L581.

39. Lavietes MH, Rochester DF. Assessment of airway function during assisted ventilation. *Lung.* 1981;159:219–229. [PubMed: 7289656]

40. Gay PC, Rodarte JR, Tayyab M, Hubmayr RD. Evaluation of bronchodilator responsiveness in mechanically ventilated patients. *Am Rev Respir Dis.* 1987;136:880–885. [PubMed: 3310772]

41. Otis AB, McKerrow CB, Bartlett RA, et al. Mechanical factors in distribution of pulmonary ventilation. *J Appl Physiol.* 1956;8:427–443. [PubMed: 13286206]

42. Fredberg JJ, Stamenovic D. On the imperfect elasticity of lung tissue. *J Appl Physiol.* 1989;67:2408–2419. [PubMed: 2606848]

43. Wilson TA, Fredberg JJ, Rodarte JR, Hyatt RE. Interdependence of regional expiratory flow. *J Appl Physiol.* 1985;59:1924–1928. [PubMed: 4077799]

44. Marini JJ, Rodriguez RM, Lamb V. The inspiratory workload of patient-initiated mechanical ventilation. *Am Rev Respir Dis.* 1986;134:902–909. [PubMed: 3096177]

45. Prechter GC, Nelson SB, Hubmayr RD. The ventilatory recruitment threshold for carbon dioxide. *Am Rev Respir Dis.* 1990;141:758–764. [PubMed: 2106815]

46. Bernard GR, Artigas A, Brigham KL, et al. The American-European Consensus Conference on ARDS. Definitions, mechanisms, relevant outcomes, and clinical trial coordination. *Am J Respir Crit Care Med.* 1994;149:818–824. [PubMed: 7509706]

47. Levy MM, Fink MP, Marshall JC, et al. 2001 SCCM/ESICM/ACCP/ATS/SIS International Sepsis Definitions Conference. *Crit Care Med.* 2003;31:1250–1256. [PubMed: 12682500]

48. Wood LD, Prewitt RM. Cardiovascular management in acute hypoxemic respiratory failure. *Am J Cardiol.* 1981;47:963–972. [PubMed: 7010980]

49. Bhandari V. Molecular mechanisms of hyperoxia-induced acute lung injury. *Front Biosci.* 2008;13:6653–6661. [PubMed: 18508685]

50. Dreyfuss D, Saumon G. Ventilator-induced lung injury: lessons from experimental studies. *Am J Respir Crit Care Med.* 1998;157:294–323. [PubMed: 9445314]

51. Weg JG, Anzueto A, Balk RA, et al. The relation of pneumothorax and other air leaks to mortality in the acute respiratory distress syndrome. *N Engl J Med.* 1998;338:341–346. [PubMed: 9449726]

52. Ryrfeldt A. Drug-induced inflammatory responses to the lung. *Toxicol Lett.* 2000;112–113:171–176. [PubMed: 10720728]

53. Maunder RJ, Shuman WP, McHugh JW, et al. Preservation of normal lung regions in the adult respiratory distress syndrome. Analysis by computed tomography. *JAMA.* 1986;255:2463–2465. [PubMed: 3701964]

54. Gattinoni L, Pesenti A, Avalli L, et al. Pressure-volume curve of total respiratory system in acute respiratory failure. Computed tomographic scan study. *Am Rev Respir Dis.* 1987;136:730–736. [PubMed: 3307572]

55. Jones AT, Hansell DM, Evans TW. Pulmonary perfusion in supine and prone positions: an electron-beam computed tomography study. *J Appl Physiol.* 2001;90:1342–1348. [PubMed: 11247933]

56. Hubmayr RD. Perspective on lung injury and recruitment: a skeptical look at the opening and collapse story. *Am J Respir Crit Care Med.* 2002;165:1647–1653. [PubMed: 12070067]

57. Bilek AM, Dee KC, Gaver DP 3rd. Mechanisms of surface-tension-induced epithelial cell damage in a model of pulmonary airway reopening. *J Appl Physiol.* 2003;94:770–783. [PubMed: 12433851]

58. Muscedere JG, Mullen JB, Gan K, Slutsky AS. Tidal ventilation at low airway pressures can augment lung injury. *Am J Respir Crit Care Med.* 1994;149:1327–1334. [PubMed: 8173774]

59. Morales MM, Pires-Neto RC, Inforsato N, et al. Small airway remodeling in acute respiratory distress syndrome: a study in autopsy lung tissue. *Crit Care.* 2011;15:R4.

60. Oeckler RA, Lee WY, Park MG, et al. Determinants of plasma membrane wounding by deforming stress. *Am J Physiol Lung Cell Mol Physiol.* 2010;299:L826–L833.

61. Huh D, Fujioka H, Tung YC, et al. Acoustically detectable cellular-level lung injury induced by fluid mechanical stresses in microfluidic airway systems. *Proc Natl Acad Sci U S A.* 2007;104:18886–18891. [PubMed: 18006663]

62. Plataki M, Hubmayr RD. The physical basis of ventilator-induced lung injury. *Expert Rev Respir Med.* 2010;4:373–385. [PubMed: 20524920]

63. Richard JC, Maggiore SM, Jonson B, et al. Influence of tidal volume on alveolar recruitment. Respective role of PEEP and a recruitment maneuver. *Am J Respir Crit Care Med.* 2001;163:1609–1613. [PubMed: 11401882]

64. Malbouisson LM, Muller JC, Constantin JM, et al. Computed tomography assessment of positive end-expiratory pressure-induced alveolar recruitment in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 2001;163:1444–1450. [PubMed: 11371416]

65. Lim SC, Adams AB, Simonson DA, et al. Intercomparison of recruitment maneuver efficacy in three models of acute lung injury. *Crit Care Med.* 2004;32:2371–2377. [PubMed: 15599138]

66. Brower RG, Morris A, MacIntyre N, et al. Effects of recruitment maneuvers in patients with acute lung injury and acute respiratory distress syndrome ventilated with high positive end-expiratory pressure. *Crit Care Med.* 2003;31:2592–2597. [PubMed: 14605529]

67. Oczenski W, Hormann C, Keller C, et al. Recruitment maneuvers after a positive end-expiratory pressure trial do not induce sustained effects in early adult respiratory distress syndrome. *Anesthesiology.* 2004;101:620–625. [PubMed: 15329586]

68. Pelosi P, Cadringher P, Bottino N, et al. Sigh in acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 1999;159:872–880. [PubMed: 10051265]

69. Villagra A, Ochagavia A, Vatua S, et al. Recruitment maneuvers during lung protective ventilation in acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 2002;165:165–170. [PubMed: 11790648]

70. Broseghini C, Brandolese R, Poggi R, et al. Respiratory resistance and intrinsic positive end-expiratory pressure (PEEPi) in patients with the adult respiratory distress syndrome (ARDS). *Eur Respir J.* 1988;1:726–731. [PubMed: 3069488]

71. Terragni PP, Antonelli M, Fumagalli R, et al. Early vs late tracheotomy for prevention of pneumonia in mechanically ventilated adult ICU patients: a randomized controlled trial. *JAMA.* 2010;303:1483–1489. [PubMed: 20407057]

72. Bull TM, Clark B, McFann K, Moss M. Pulmonary vascular dysfunction is associated with poor outcomes in patients with acute lung injury. *Am J Respir Crit Care Med.* 2010;182:1123–1128. [PubMed: 20558628]

73. Meade MO, Cook DJ, Guyatt GH, et al. Ventilation strategy using low tidal volumes, recruitment maneuvers, and high positive end-expiratory pressure for acute lung injury and acute respiratory distress syndrome: a randomized controlled trial. *JAMA.* 2008;299:637–645. [PubMed: 18270352]

74. Mercat A, Richard JC, Vielle B, et al. Positive end-expiratory pressure setting in adults with acute lung injury and acute respiratory distress syndrome: a randomized controlled trial. *JAMA.* 2008;299:646–655. [PubMed: 18270353]

75. Brower RG, Lanken PN, MacIntyre N, et al. Higher versus lower positive end-expiratory pressures in patients with the acute respiratory distress syndrome. *N Engl J Med.* 2004;351:327–336. [PubMed: 15269312]

76. Gattinoni L, Caironi P, Cressoni M, et al. Lung recruitment in patients with the acute respiratory distress syndrome. *N Engl J Med.* 2006;354:1775–1786. [PubMed: 16641394]

77. Rouby JJ, Puybasset L, Nieszkowska A, Lu Q. Acute respiratory distress syndrome: lessons from computed tomography of the whole lung. *Crit Care Med.* 2003;31:S285–S295.

78. Victorino JA, Borges JB, Okamoto VN, et al. Imbalances in regional lung ventilation: a validation study on electrical impedance tomography. *Am J Respir Crit Care Med.* 2004;169:791–800. [PubMed: 14693669]

79. Jonson B, Richard JC, Straus C, et al. Pressure-volume curves and compliance in acute lung injury: evidence of recruitment above the lower inflection point. *Am J Respir Crit Care Med.* 1999;159:1172–1178. [PubMed: 10194162]

80. Maggiore SM, Jonson B, Richard JC, et al. Alveolar derecruitment at decremental positive end-expiratory pressure levels in acute lung injury: comparison with the lower inflection point, oxygenation, and compliance. *Am J Respir Crit Care Med.* 2001;164:795–801. [PubMed: 11549535]

81. Grasso S, Terragni P, Mascia L, et al. Airway pressure-time curve profile (stress index) detects tidal recruitment/hyperinflation in experimental acute lung injury. *Crit Care Med.* 2004;32:1018–1027. [PubMed: 15071395]

82. Talmor D, Sarge T, O'Donnell CR, et al. Esophageal and transpulmonary pressures in acute respiratory failure. *Crit Care Med.* 2006;34:1389–1394. [PubMed: 16540960]

83. Baumgardner JE, Markstaller K, Pfeiffer B, et al. Effects of respiratory rate, plateau pressure, and positive end-expiratory pressure on Paw_2 oscillations after saline lavage. *Am J Respir Crit Care Med.* 2002;166:1556–1562. [PubMed: 12406831]

84. Malbrain ML, Chiumello D, Pelosi P, et al. Incidence and prognosis of intraabdominal hypertension in a mixed population of critically ill patients: a multiple-center epidemiological study. *Crit Care Med.* 2005;33:315–322. [PubMed: 15699833]

85. Mattingley JS, Holets SR, Oeckler RA, et al. Sizing the lung of mechanically ventilated patients. *Crit Care.* 2011;15:R60.

86. Malbrain ML. Abdominal pressure in the critically ill: measurement and clinical relevance. *Intensive Care Med.* 1999;25:1453–1458. [PubMed: 10702030]

87. Pelosi P, Croci M, Ravagnan I, et al. The effects of body mass on lung volumes, respiratory mechanics, and gas exchange during general anesthesia. *Anesth Analg.* 1998;87:654–660. [PubMed: 9728848]

88. Malo J, Ali J, Wood LD. How does positive end-expiratory pressure reduce intrapulmonary shunt in canine pulmonary edema? *J Appl Physiol.* 1984;57:1002–1010. [PubMed: 6389451]

89. Gattinoni L, Pelosi P, Suter PM, et al. Acute respiratory distress syndrome caused by pulmonary and extrapulmonary disease. Different syndromes? *Am J Respir Crit Care Med.* 1998;158:3–11. [PubMed: 9655699]

90. Squadrone V, Coha M, Cerutti E, et al. Continuous positive airway pressure for treatment of postoperative hypoxemia: a randomized controlled trial. *JAMA.* 2005;293:589–595. [PubMed: 15687314]

91. Rouby JJ, Lu Q, Goldstein I. Selecting the right level of positive end-expiratory pressure in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 2002;165:1182–1186. [PubMed: 11956065]

92. Fernandez-Perez ER, Yilmaz M, Jenad H, et al. Ventilator settings and outcome of respiratory failure in chronic interstitial lung disease. *Chest.* 2008;133:1113–1119. [PubMed: 17989156]

93. Talmor D, Sarge T, Malhotra A, et al. Mechanical ventilation guided by esophageal pressure in acute lung injury. *N Engl J Med.* 2008;359:2095–2104. [PubMed: 19001507]

94. Mead J, Gaensler EA. Esophageal and pleural pressures in man, upright and supine. *J Appl Physiol.* 1959;14:81–83. [PubMed: 13630830]

95. Plataki M, Hubmayr RD. Should mechanical ventilation be guided by esophageal pressure measurements? *Curr Opin Crit Care.* 2011;17:275–280. [PubMed: 21326094]

96. Rivers EP, Ander DS, Powell D. Central venous oxygen saturation monitoring in the critically ill patient. *Curr Opin Crit Care.* 2001;7:204–211. [PubMed: 11436529]

97. Cain SM. Oxygen supply dependency in the critically ill—a continuing conundrum. *Adv Exp Med Biol.* 1992;317:35–45. [PubMed: 1288144]

98. Hood VL, Tannen RL. Protection of acid-base balance by pH regulation of acid production. *N Engl J Med.* 1998;339:819–826. [PubMed: 9738091]

99. Hickling KG, Joyce C. Permissive hypercapnia in ARDS and its effect on tissue oxygenation. *Acta Anaesthesiol Scand Suppl.* 1995;107:201–208. [PubMed: 8599278]

100. Beydon L, Uttman L, Rawal R, Jonson B. Effects of positive end-expiratory pressure on dead space and its partitions in acute lung injury. *Intensive Care Med.* 2002;28:1239–1245. [PubMed: 12209271]

101. Lim SC, Adams AB, Simonson DA, et al. Transient hemodynamic effects of recruitment maneuvers in three experimental models of acute lung injury. *Crit Care Med.* 2004;32:2378–2384. [PubMed: 15599139]

102. Pinsky MR. Recent advances in the clinical application of heart-lung interactions. *Curr Opin Crit Care.* 2002;8:26–31. [PubMed: 12205403]

103. Horie T, Hildebrandt J. Volume history, static equilibrium, and dynamic compliance of excised cat lung. *J Appl Physiol.* 1972;33:105–112. [PubMed: 5037393]

104. Caples SM, Hubmayr RD. Respiratory monitoring tools in the intensive care unit. *Curr Opin Crit Care.* 2003;9:230–235. [PubMed: 12771676]

105. Gajic O, Dara SI, Mendez JL, et al. Ventilator-associated lung injury in patients without acute lung injury at the onset of mechanical ventilation. *Crit Care Med.* 2004;32:1817–1824. [PubMed: 15343007]

106. Vlahakis NE, Hubmayr RD. Cellular stress failure in ventilator-injured lungs. *Am J Respir Crit Care Med.* 2005;171:1328–1342. [PubMed: 15695492]

107. Nieszkowska A, Lu Q, Vieira S, et al. Incidence and regional distribution of lung overinflation during mechanical ventilation with positive end-expiratory pressure. *Crit Care Med.* 2004;32:1496–1503. [PubMed: 15241094]

108. Veldhuizen RA, Welk B, Harbottle R, et al. Mechanical ventilation of isolated rat lungs changes the structure and biophysical properties of surfactant. *J Appl Physiol.* 2002;92:1169–1175. [PubMed: 11842055]

109. Veldhuizen RA, Yao LJ, Lewis JF. An examination of the different variables affecting surfactant aggregate conversion in vitro. *Exp Lung Res.* 1999;25:127–141. [PubMed: 10188107]

110. Mead J, Takishima T, Leith D. Stress distribution in lungs: a model of pulmonary elasticity. *J Appl Physiol.* 1970;28:596–608. [PubMed: 5442255]

111. Martynowicz MA, Minor TA, Walters BJ, Hubmayr RD. Regional expansion of oleic acid-injured lungs. *Am J Respir Crit Care Med.* 1999;160:250–258. [PubMed: 10390408]

112. Gattinoni L, Pesenti A. The concept of "baby lung." *Intensive Care Med.* 2005;31:776–784. [PubMed: 15812622]

113. Chiumello D, Carlesso E, Cadringher P, et al. Lung stress and strain during mechanical ventilation for acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 2008;178:346–355. [PubMed: 18451319]

114. Crapo RO, Morris AH, Clayton PD, Nixon CR. Lung volumes in healthy nonsmoking adults. *Bull Eur Physiopathol Respir.* 1982;18:419–425. [PubMed: 7074238]

115. Standardized lung function testing. Official statement of the European Respiratory Society. *Eur Respir J Suppl.* 1993;16:1–100.

116. Protti A, Cressoni M, Santini A, et al. Lung stress and strain during mechanical ventilation: any safe threshold? *Am J Respir Crit Care Med.* 2011;183:1354–1362. [PubMed: 21297069]

117. Terragni PP, Rosboch G, Tealdi A, et al. Tidal hyperinflation during low tidal volume ventilation in acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 2007;175:160–166. [PubMed: 17038660]

118. Loring SH, O'Donnell CR, Behazin N, et al. Esophageal pressures in acute lung injury: do they represent artifact or useful information about transpulmonary pressure, chest wall mechanics, and lung stress? *J Appl Physiol.* 2010;108:515–522. [PubMed: 20019160]

119. Gattinoni L, Pesenti A, Bombino M, et al. Relationships between lung computed tomographic density, gas exchange, and PEEP in acute respiratory failure. *Anesthesiology.* 1988;69:824–832. [PubMed: 3057937]

120. Eichacker PQ, Gerstenberger EP, Banks SM, et al. Meta-analysis of acute lung injury and acute respiratory distress syndrome trials testing low tidal volumes. *Am J Respir Crit Care Med.* 2002;166:1510–1514. [PubMed: 12406836]

121. Rehder K, Abboud N, Rodarte JR, Hyatt RE. Positive airway pressure and vertical transpulmonary pressure gradient in man. *J Appl Physiol.* 1975;38:896–899. [PubMed: 1126900]

122. Loring SH, Weiss JW. Plateau pressures in the ARDSnet protocol: cause of injury or indication of disease? *Am J Respir Crit Care Med.* 2007;176:99–100; author reply 100–101. [PubMed: 17586764]

123. Mascheroni D, Kolobow T, Fumagalli R, et al. Acute respiratory failure following pharmacologically induced hyperventilation: an experimental animal study. *Intensive Care Med.* 1988;15:8–14. [PubMed: 3230208]

124. Hager DN, Krishnan JA, Hayden DL, Brower RG. Tidal volume reduction in patients with acute lung injury when plateau pressures are not high. *Am J Respir Crit Care Med.* 2005;172:1241–1245. [PubMed: 16081547]

125. Putensen C, Mutz NJ, Putensen-Himmer G, Zinserling J. Spontaneous breathing during ventilatory support improves ventilation-perfusion distributions in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 1999;159:1241–1248. [PubMed: 10194172]

126. Putensen C, Hering R, Muders T, Wrigge H. Assisted breathing is better in acute respiratory failure. *Curr Opin Crit Care.* 2005;11:63–68. [PubMed: 15659947]

127. Hubmayr RD, Rodarte JR, Walters BJ, Tonelli FM. Regional ventilation during spontaneous breathing and mechanical ventilation in dogs. *J Appl Physiol.* 1987;63:2467–2475. [PubMed: 3436878]

128. Reber A, Nylund U, Hedenstierna G. Position and shape of the diaphragm: implications for atelectasis formation. *Anaesthesia.* 1998;53:1054–1061. [PubMed: 10023273]

129. Wrigge H, Uhlig U, Zinserling J, et al. The effects of different ventilatory settings on pulmonary and systemic inflammatory responses during major surgery. *Anesth Analg.* 2004;98:775–781, table of contents. [PubMed: 14980936]

130. Beck-Schimmer B, Schimmer RC. Perioperative tidal volume and intra-operative open lung strategy in healthy lungs: where are we going? *Best Pract Res Clin Anaesthesiol.* 2010;24:199–210. [PubMed: 20608557]

131. Determann RM, Royakkers A, Wolthuis EK, et al. Ventilation with lower tidal volumes as compared with conventional tidal volumes for patients without acute lung injury: a preventive randomized controlled trial. *Crit Care.* 2010;14:R1.

132. Cole AG, Weller SF, Sykes MK. Inverse ratio ventilation compared with PEEP in adult respiratory failure. *Intensive Care Med.* 1984;10:227–232. [PubMed: 6386921]

133. Yang SC, Yang SP. Effects of inspiratory flow waveforms on lung mechanics, gas exchange, and respiratory metabolism in COPD patients during mechanical ventilation. *Chest.* 2002;122:2096–2104. [PubMed: 12475853]

134. Laghi F, Segal J, Choe WK, Tobin MJ. Effect of imposed inflation time on respiratory frequency and hyperinflation in patients with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med.* 2001;163:1365–1370. [PubMed: 11371402]

135. Kallet RH, Alonso JA, Pittet JF, Matthay MA. Prognostic value of the pulmonary dead-space fraction during the first 6 days of acute respiratory distress syndrome. *Respir Care.* 2004;49:1008–1014. [PubMed: 15329171]

136. Darioli R, Perret C. Mechanical controlled hypoventilation in status asthmaticus. *Am Rev Respir Dis.* 1984;129:385–387. [PubMed: 6703497]

137. Hickling KG, Henderson SJ, Jackson R. Low mortality associated with low volume pressure limited ventilation with permissive hypercapnia in severe adult respiratory distress syndrome. *Intensive Care Med.* 1990;16:372–377. [PubMed: 2246418]

138. Papazian L, Forel JM, Gacouin A, et al. Neuromuscular blockers in early acute respiratory distress syndrome. *N Engl J Med.* 2010; 363:1107–1116. [PubMed: 20843245]

139. Broccard AF, Hotchkiss JR, Vannay C, et al. Protective effects of hypercapnic acidosis on ventilator-induced lung injury. *Am J Respir Crit Care Med.* 2001;164:802–806. [PubMed: 11549536]

140. Laffey JG, Tanaka M, Engelberts D, et al. Therapeutic hypercapnia reduces pulmonary and systemic injury following in vivo lung reperfusion. *Am J Respir Crit Care Med.* 2000;162:2287–2294. [PubMed: 11112153]

141. Doerr CH, Gajic O, Berrios JC, et al. Hypercapnic acidosis impairs plasma membrane wound resealing in ventilator-injured lungs. *Am J Respir Crit Care Med.* 2005;171:1371–1377. [PubMed: 15695495]

142. Rossi N, Musch G, Sangalli F, et al. Reverse-thrust ventilation in hypercapnic patients with acute respiratory distress syndrome. Acute physiological effects. *Am J Respir Crit Care Med.* 2000;162:363–368. [PubMed: 10934054]

143. Richecoeur J, Lu Q, Vieira SR, et al. Expiratory washout versus optimization of mechanical ventilation during permissive hypercapnia in patients with severe acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 1999;160:77–85. [PubMed: 10390383]

144. Peek GJ, Mugford M, Tiruvoipati R, et al. Efficacy and economic assessment of conventional ventilatory support versus extracorporeal membrane oxygenation for severe adult respiratory failure (CESAR): a multicentre randomised controlled trial. *Lancet.* 2009;374:1351–1363. [PubMed: 19762075]

145. Hubmayr RD, Farmer JC. Should we “rescue” patients with 2009 influenza A(H1N1) and lung injury from conventional mechanical ventilation? *Chest.* 2010;137:745–747. [PubMed: 20371522]

146. Reinoso MA, Gracey DR, Hubmayr RD. Interrupter mechanics of patients admitted to a chronic ventilator dependency unit. *Am Rev Respir Dis.* 1993;148:127–131. [PubMed: 8317788]

147. Pepe PE, Marini JJ. Occult positive end-expiratory pressure in mechanically ventilated patients with airflow obstruction: the auto-PEEP effect. *Am Rev Respir Dis.* 1982;126:166–170. [PubMed: 7046541]

148. Williams TJ, Tuxen DV, Scheinkestel CD, et al. Risk factors for morbidity in mechanically ventilated patients with acute severe asthma. *Am Rev Respir Dis.* 1992;146:607–615. [PubMed: 1519836]

149. Sinderby C, Spahija J, Beck J, et al. Diaphragm activation during exercise in chronic obstructive pulmonary disease. *Am J Respir Crit Care Med.* 2001;163:1637–1641. [PubMed: 11401887]

150. Cassart M, Pettiaux N, Gevenois PA, et al. Effect of chronic hyperinflation on diaphragm length and surface area. *Am J Respir Crit Care Med.* 1997;156:504–508. [PubMed: 9279231]

151. Cassart M, Hamacher J, Verbandt Y, et al. Effects of lung volume reduction surgery for emphysema on diaphragm dimensions and configuration. *Am J Respir Crit Care Med.* 2001;163:1171–1175. [PubMed: 11316655]

152. Rhoney DH, Murry KR. National survey of the use of sedating drugs, neuromuscular blocking agents, and reversal agents in the intensive care unit. *J Intensive Care Med.* 2003;18:139–145. [PubMed: 14984632]

153. Parthasarathy S, Jubran A, Tobin MJ. Cycling of inspiratory and expiratory muscle groups with the ventilator in airflow limitation. *Am J Respir Crit Care Med.* 1998;158:1471–1478. [PubMed: 9817695]

154. Weber T, Tschernich H, Sitzwohl C, et al. Tromethamine buffer modifies the depressant effect of permissive hypercapnia on myocardial contractility in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 2000;162:1361–1365. [PubMed: 11029345]

155. Tuxen DV, Lane S. The effects of ventilatory pattern on hyperinflation, airway pressures, and circulation in mechanical ventilation of patients with severe air-flow obstruction. *Am Rev Respir Dis.* 1987;136:872–879. [PubMed: 3662241]

156. Tuxen DV, Williams TJ, Scheinkestel CD, et al. Use of a measurement of pulmonary hyperinflation to control the level of mechanical ventilation in patients with acute severe asthma. *Am Rev Respir Dis.* 1992;146:1136–1142. [PubMed: 1443862]

157. Petrof BJ, Legare M, Goldberg P, et al. Continuous positive airway pressure reduces work of breathing and dyspnea during weaning from mechanical ventilation in severe chronic obstructive pulmonary disease. *Am Rev Respir Dis.* 1990;141:281–289. [PubMed: 2405757]

158. Gay PC, Rodarte JR, Hubmayr RD. The effects of positive expiratory pressure on isovolume flow and dynamic hyperinflation in patients receiving mechanical ventilation. *Am Rev Respir Dis.* 1989; 139:621–626. [PubMed: 2647006]

159. Martin JG, Shore S, Engel LA. Effect of continuous positive airway pressure on respiratory mechanics and pattern of breathing in induced asthma. *Am Rev Respir Dis.* 1982;126:812–817. [PubMed: 6216839]

160. Roussos C, Macklem PT. The respiratory muscles. *N Engl J Med.* 1982;307:786–797. [PubMed: 7050712]

161. Laghi F, Cattapan SE, Jubran A, et al. Is weaning failure caused by low-frequency fatigue of the diaphragm? *Am J Respir Crit Care Med.* 2003;167:120–127. [PubMed: 12411288]

162. Graves C, Glass L, Laporta D, et al. Respiratory phase locking during mechanical ventilation in anesthetized human subjects. *Am J Physiol.* 1986;250:R902–R909.

163. Ingrassia TS 3rd, Nelson SB, Harris CD, Hubmayr RD. Influence of sleep state on CO2 responsiveness. A study of the unloaded respiratory pump in humans. *Am Rev Respir Dis.* 1991;144:1125–1129. [PubMed: 1952443]

164. Simon PM, Zurob AS, Wies WM, et al. Entrainment of respiration in humans by periodic lung inflations. Effect of state and CO(2). *Am J Respir Crit Care Med.* 1999;160:950–960. [PubMed: 10471624]

165. Juan G, Calverley P, Talamo C, et al. Effect of carbon dioxide on diaphragmatic function in human beings. *N Engl J Med.* 1984;310:874–879. [PubMed: 6422298]

166. Nava S, Fanfulla F, Frigerio P, Navalesi P. Physiologic evaluation of 4 weeks of nocturnal nasal positive pressure ventilation in stable hypercapnic patients with chronic obstructive pulmonary disease. *Respiration.* 2001;68:573–583. [PubMed: 11786711]

167. Laffey JG, Kavanagh BP. Hypocapnia. *N Engl J Med.* 2002;347:43–53. [PubMed: 12097540]

168. Jackson JC, Girard TD, Gordon SM, et al. Long-term cognitive and psychological outcomes in the awakening and breathing controlled trial. *Am J Respir Crit Care Med.* 2010;182:183–191. [PubMed: 20299535]